

BFS Traversal

Step 1: Start at E

Legend

Complete

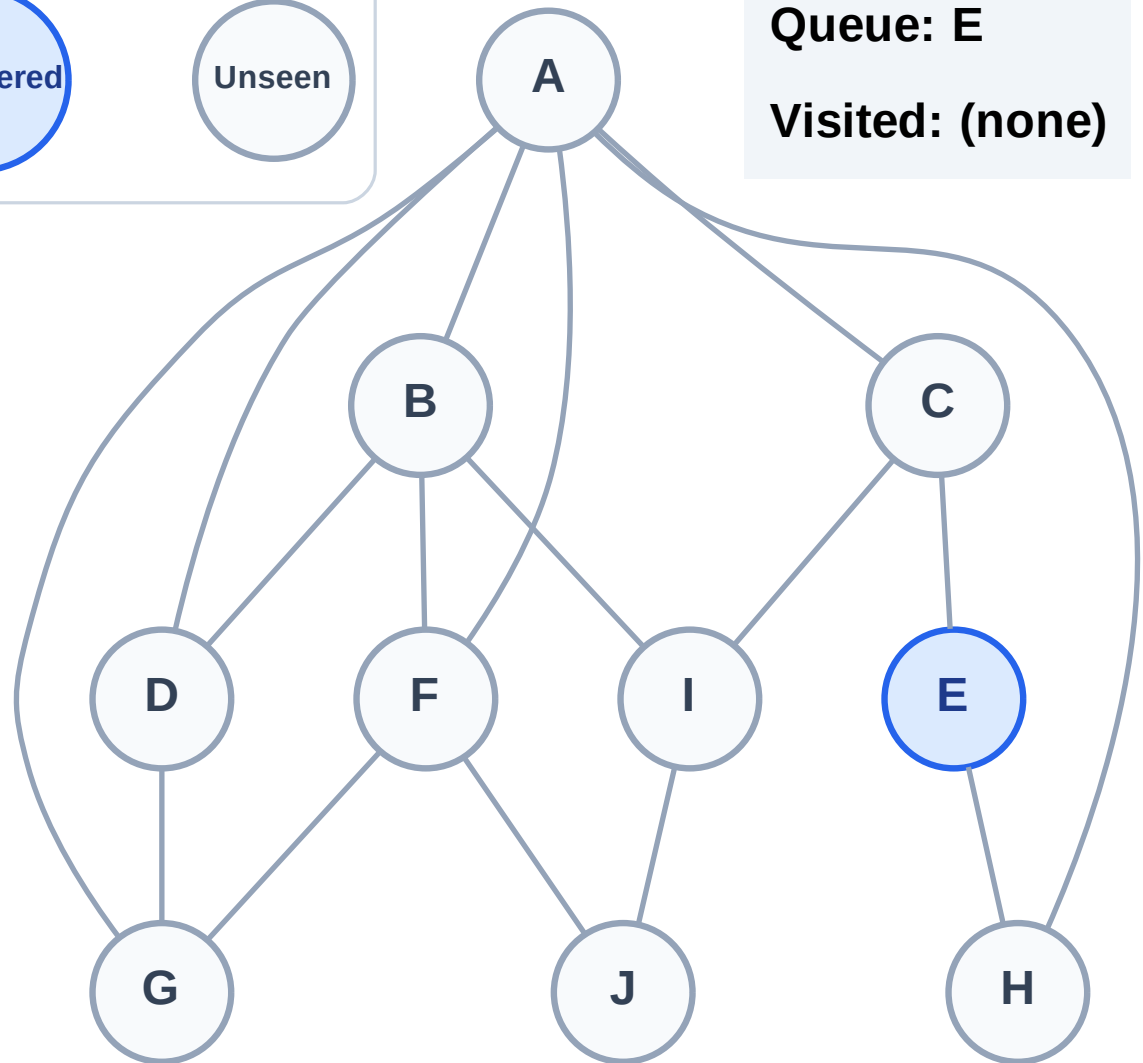
Processing

Discovered

Unseen

Queue: E

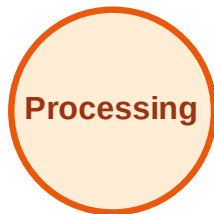
Visited: (none)



BFS Traversal

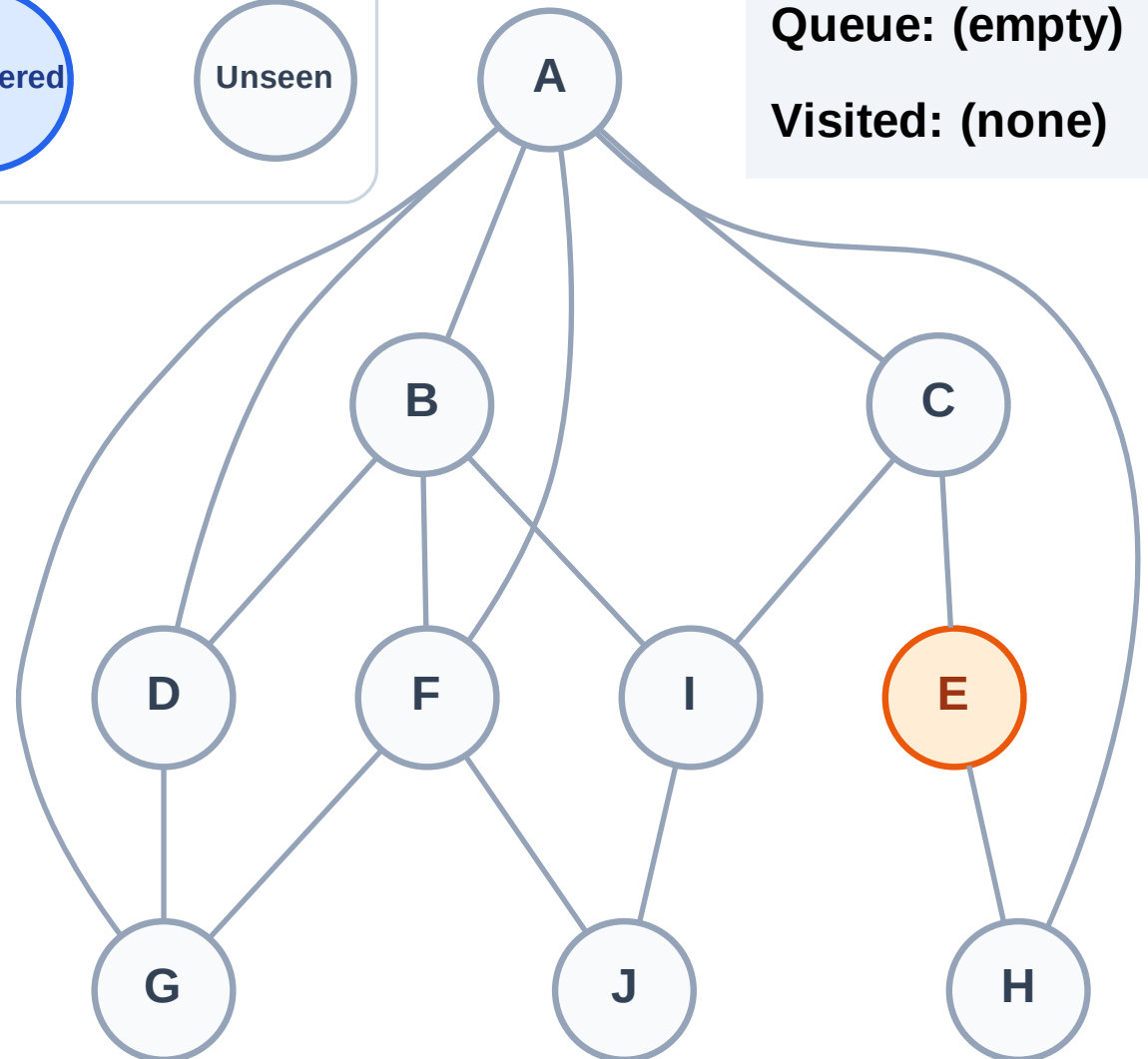
Step 2: Process node E

Legend



Queue: (empty)

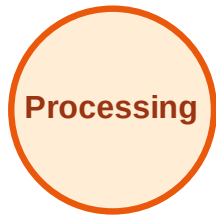
Visited: (none)



BFS Traversal

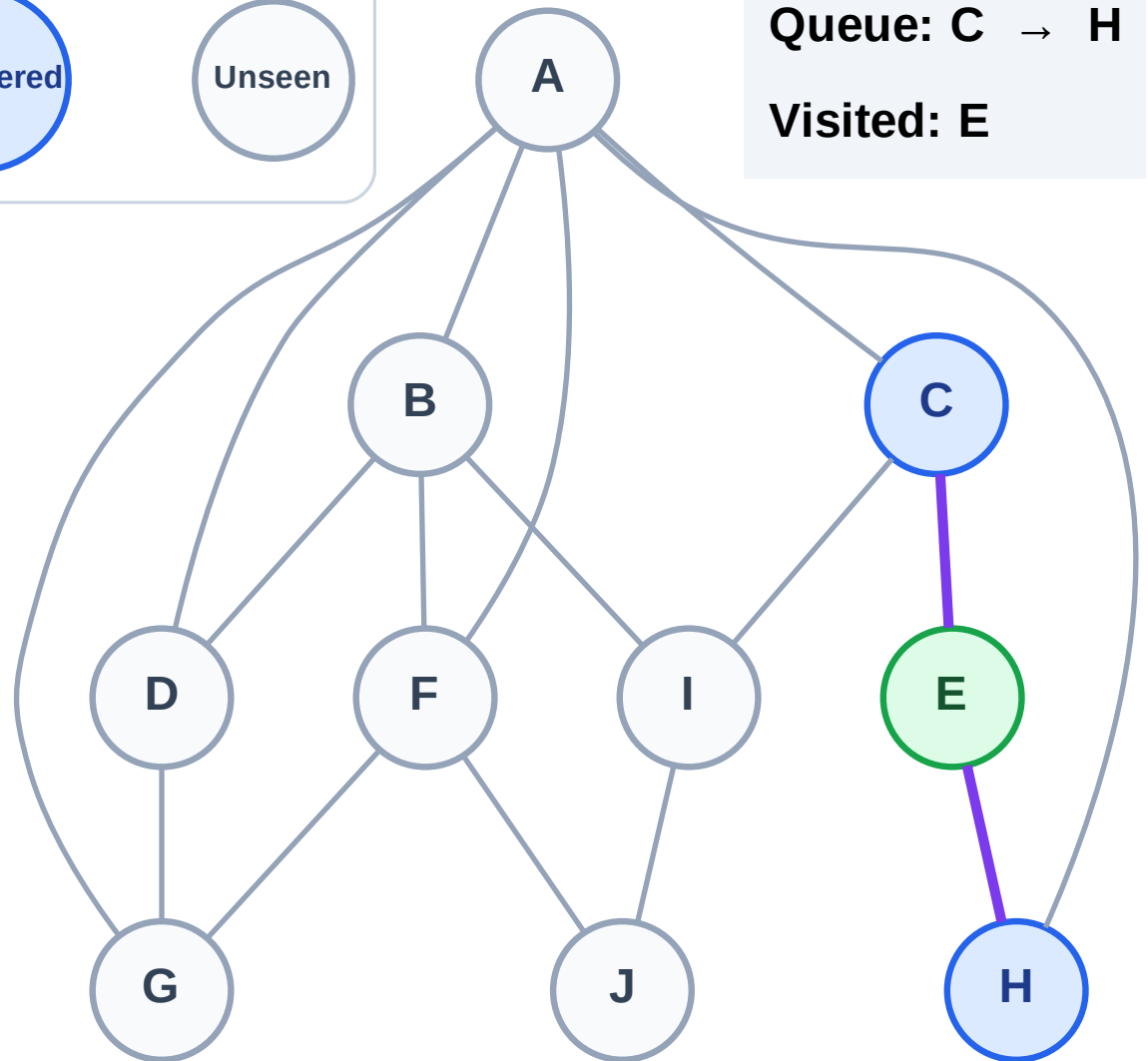
Step 3: Discover C, H from E

Legend



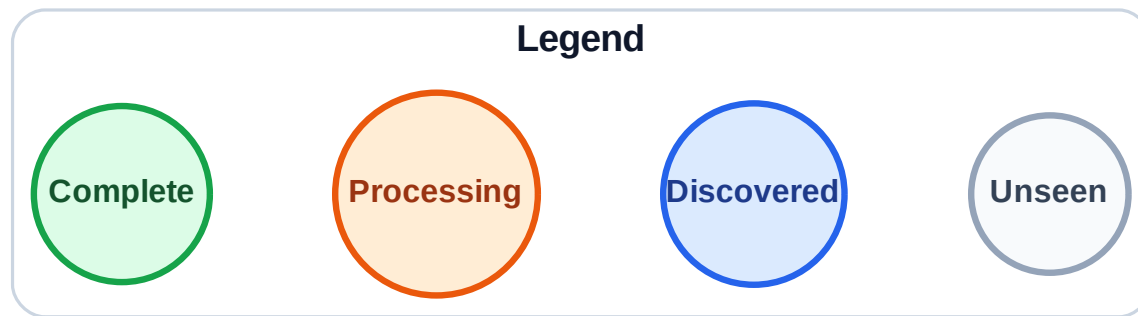
Queue: C → H

Visited: E

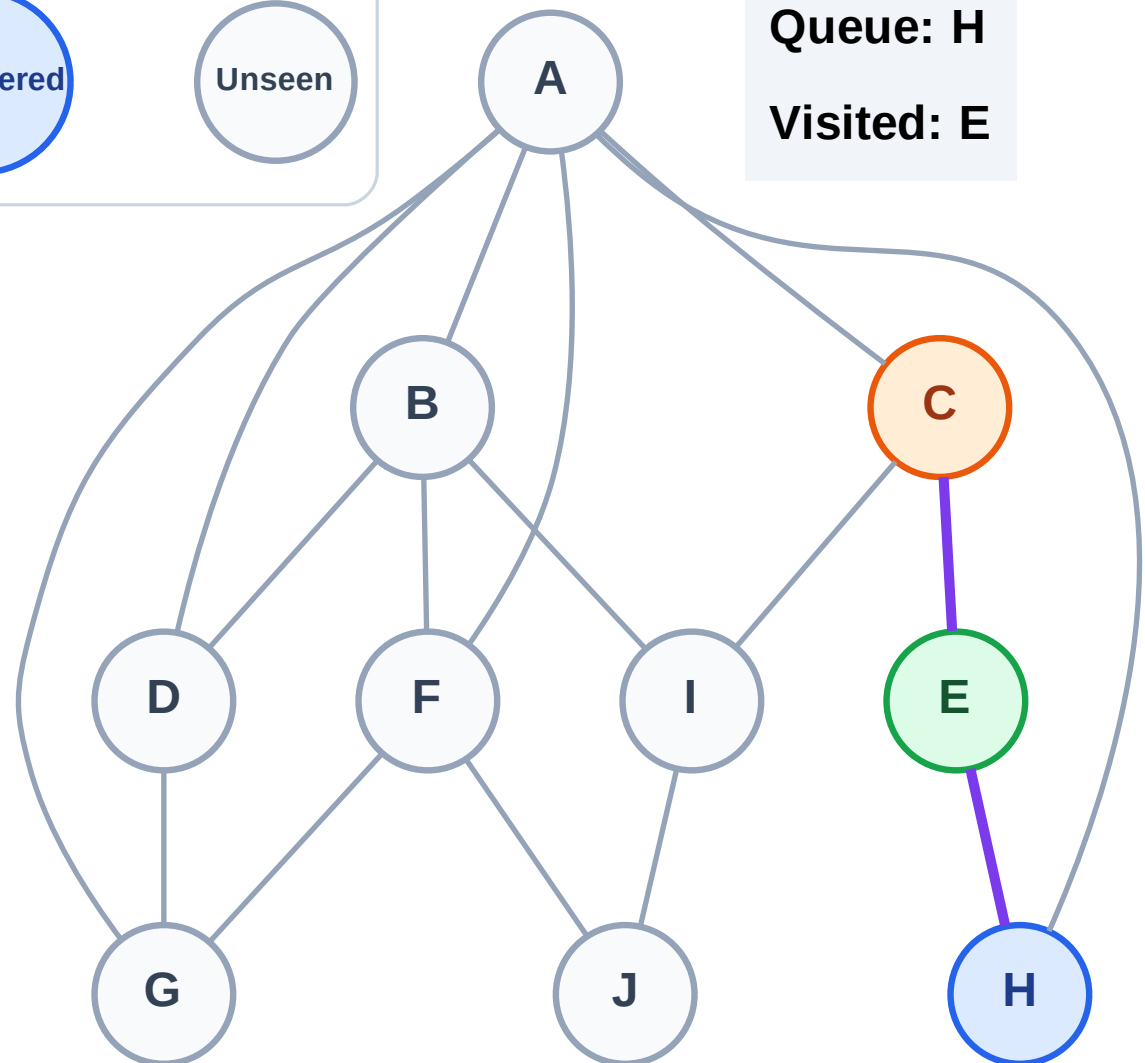


BFS Traversal

Step 4: Process node C



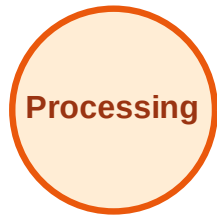
Queue: H
Visited: E



BFS Traversal

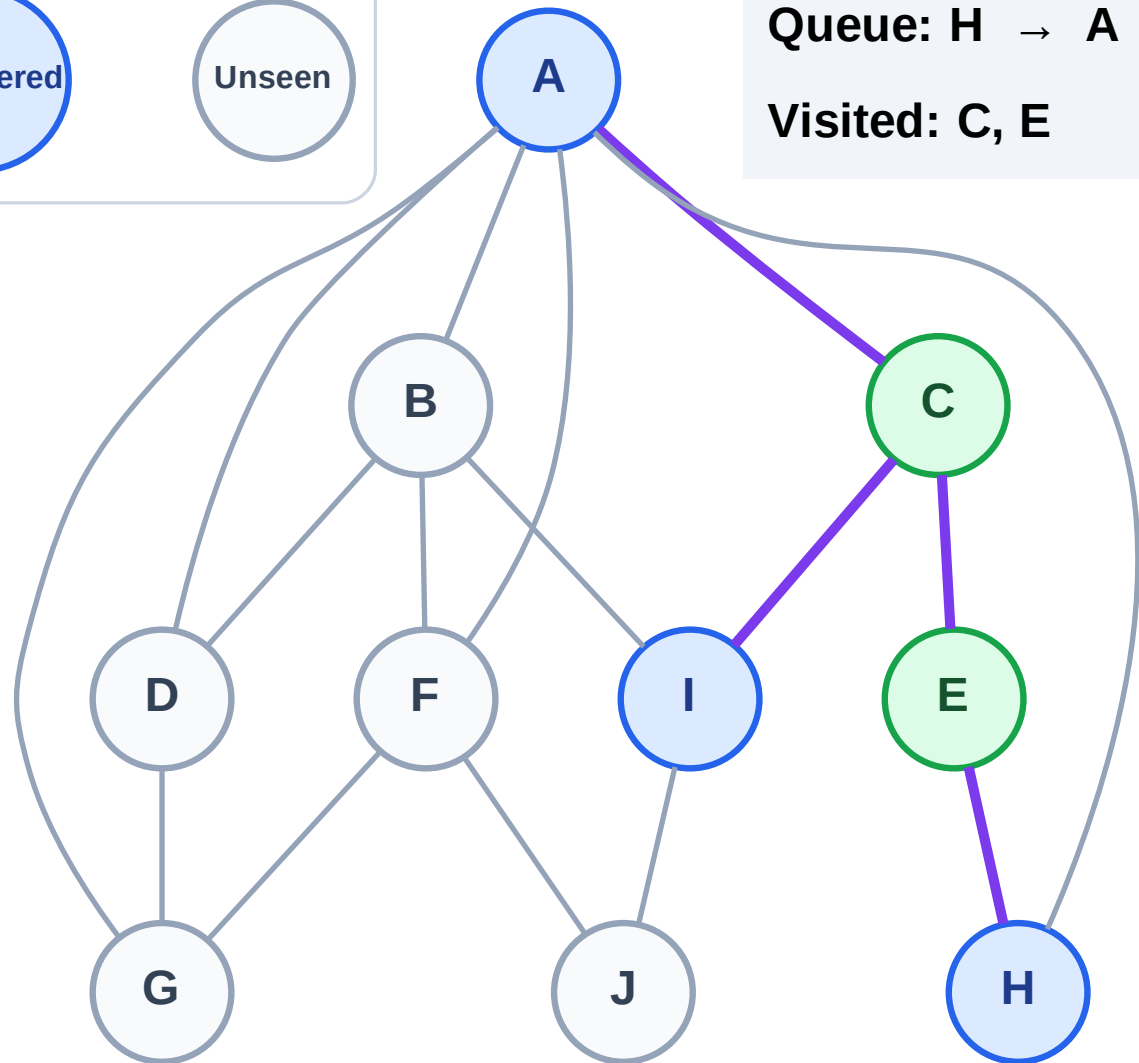
Step 5: Discover A, I from C

Legend



Queue: H → A → I

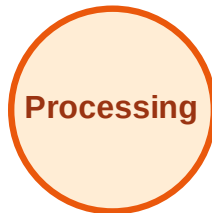
Visited: C, E



BFS Traversal

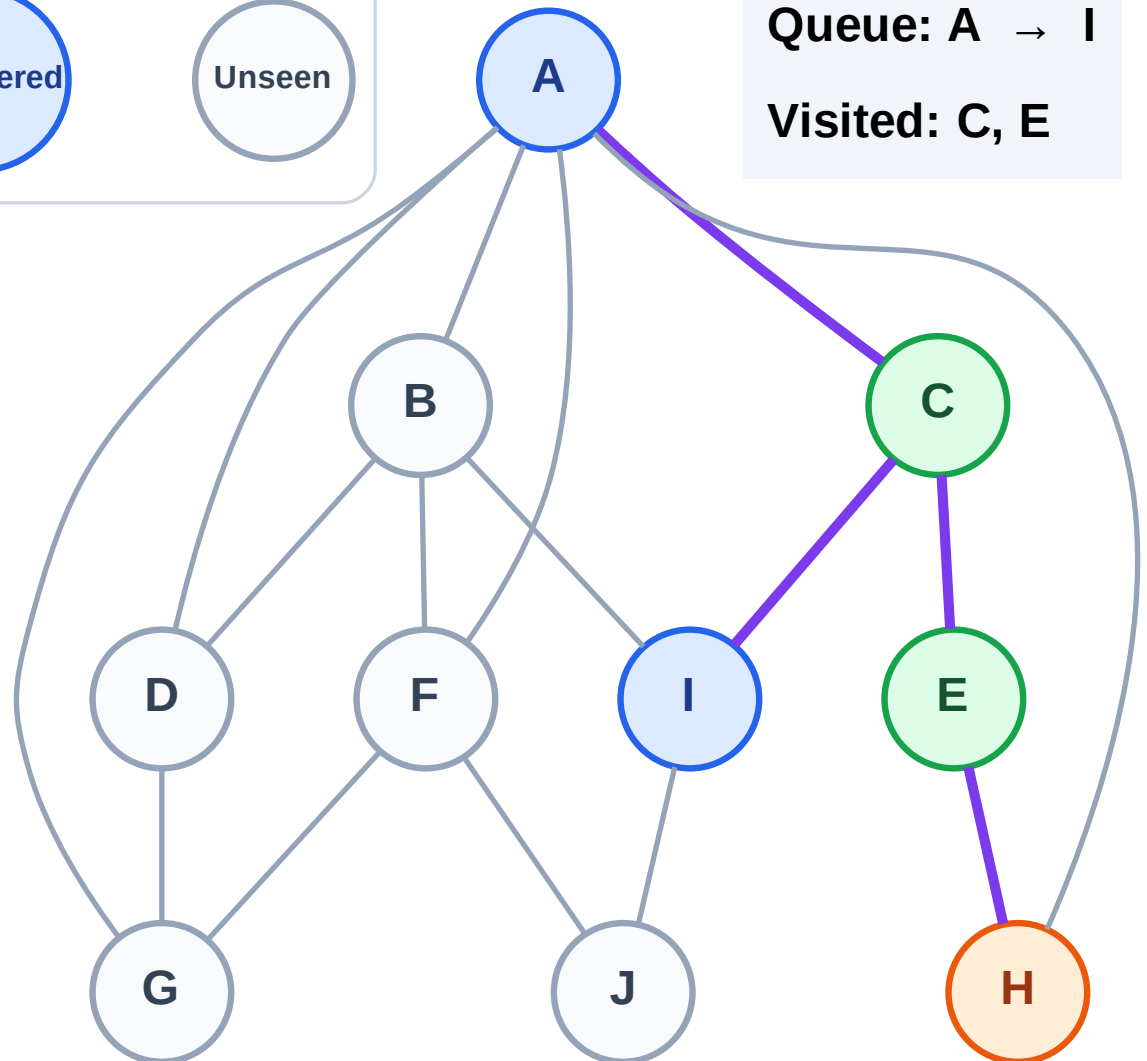
Step 6: Process node H

Legend



Queue: A → I

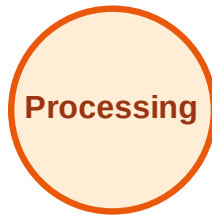
Visited: C, E



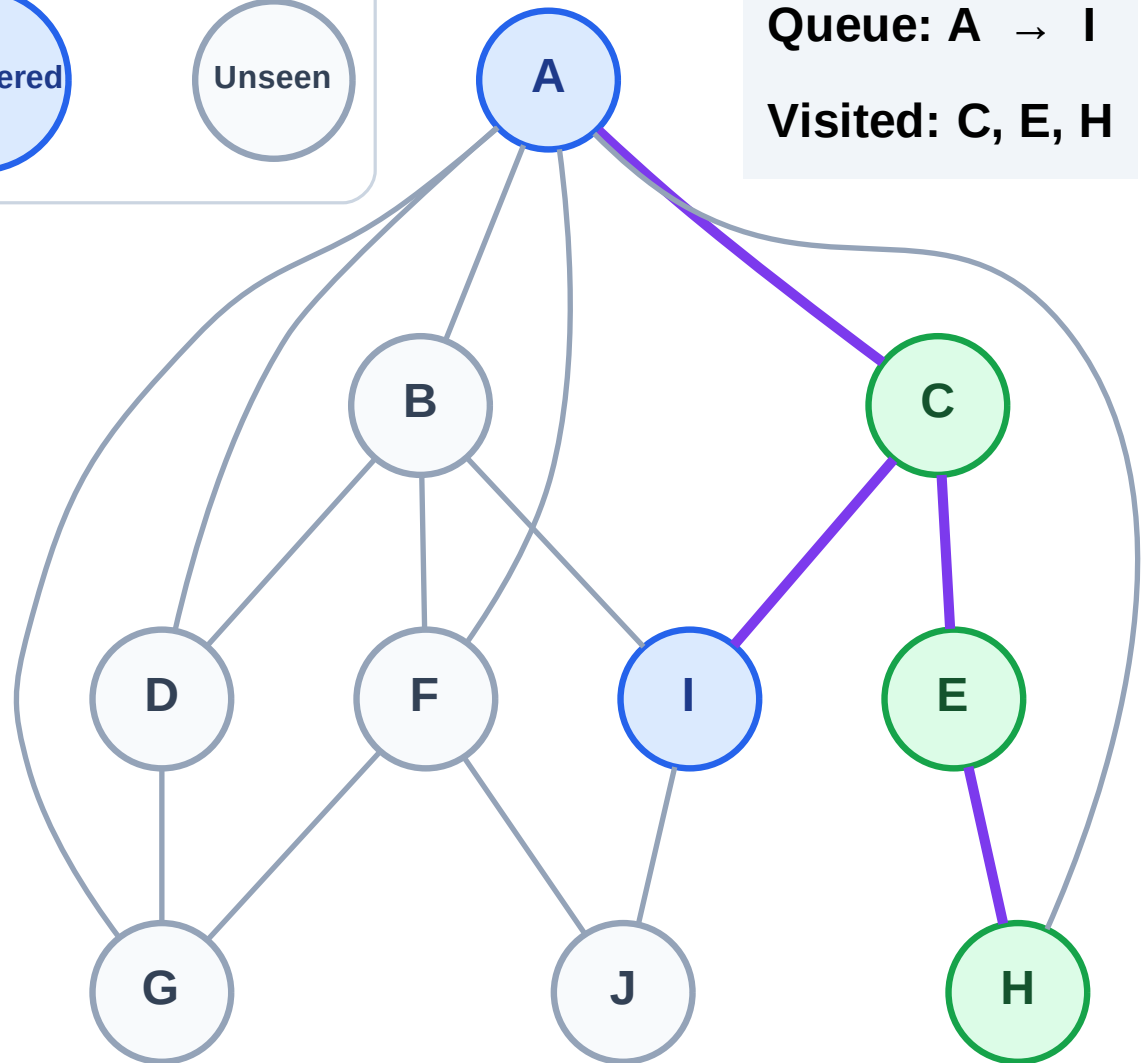
BFS Traversal

Step 7: No new neighbors from H

Legend



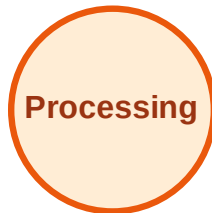
Queue: A → I
Visited: C, E, H



BFS Traversal

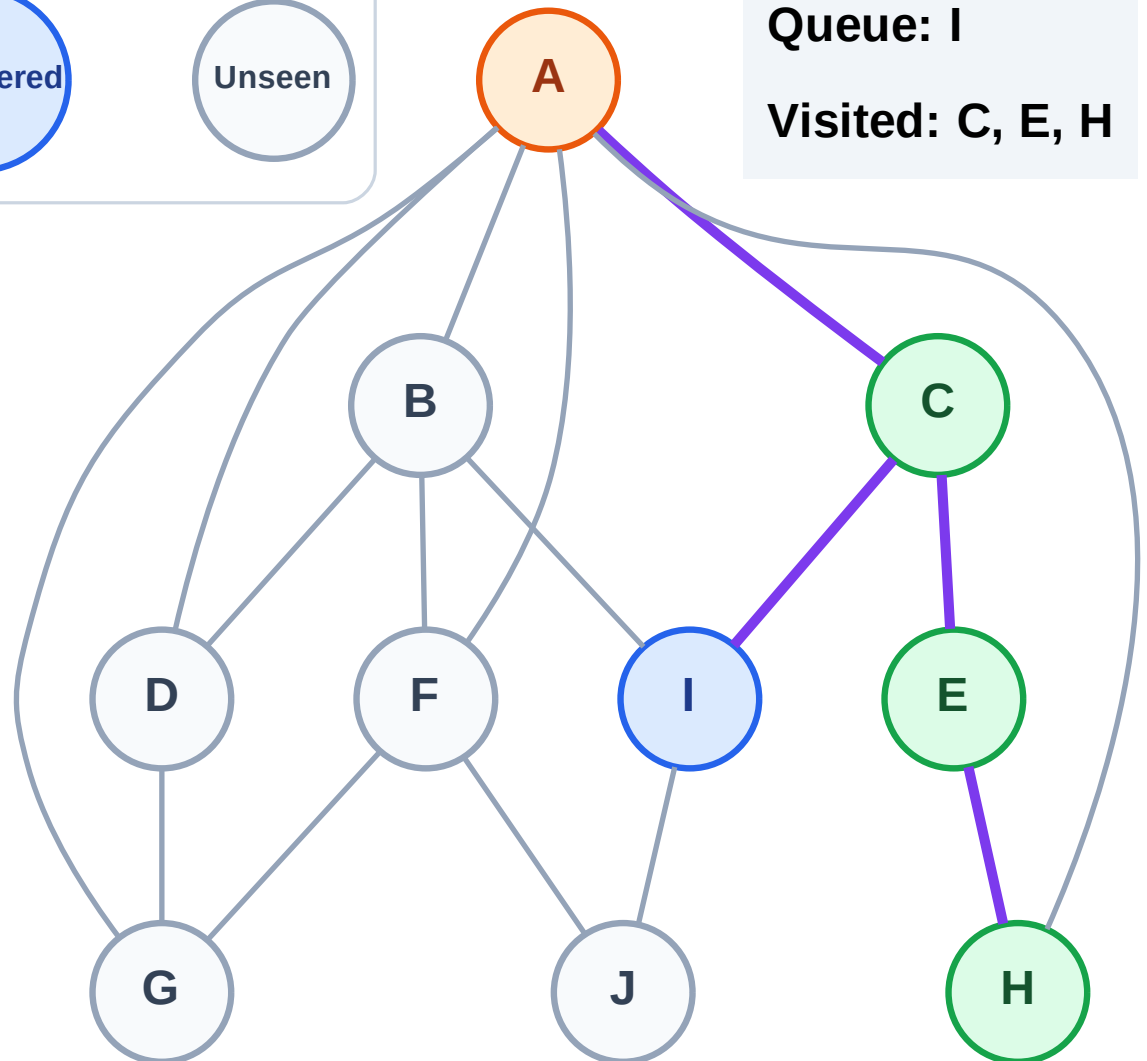
Step 8: Process node A

Legend



Queue: I

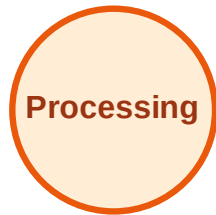
Visited: C, E, H



BFS Traversal

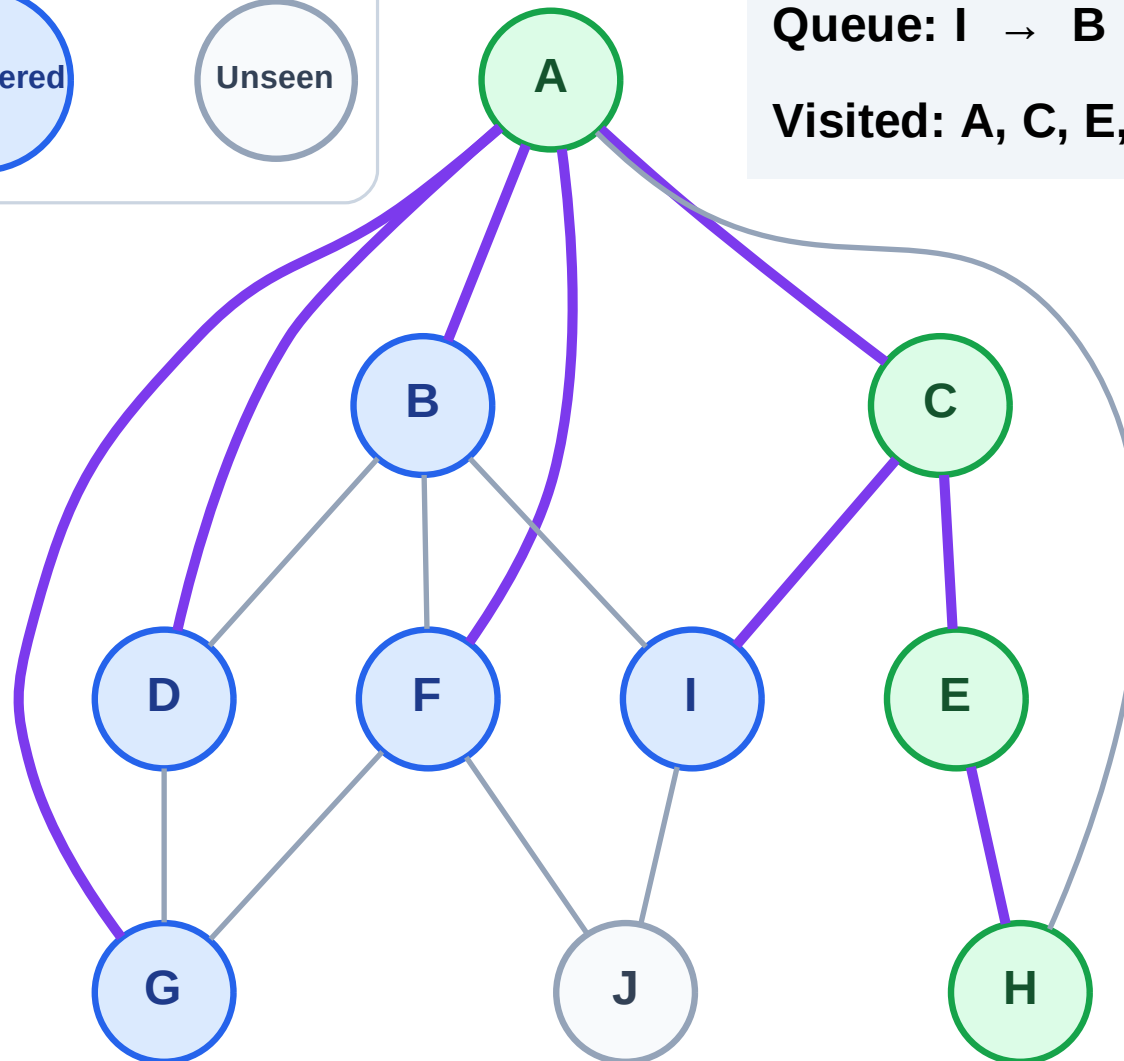
Step 9: Discover B, D, F, G from A

Legend



Queue: I → B → D → F → G

Visited: A, C, E, H



BFS Traversal

Step 10: Process node I

Legend

Complete

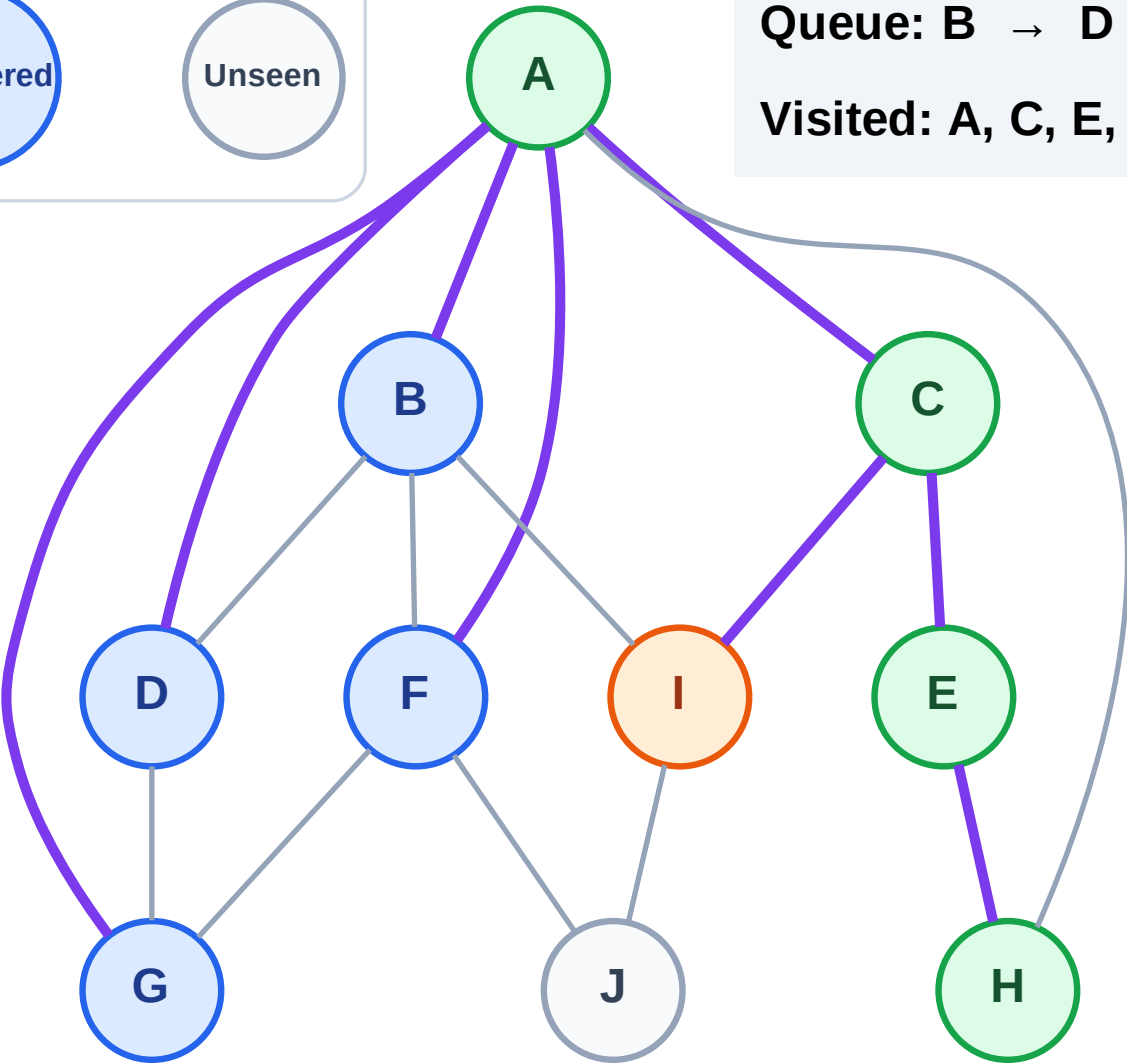
Processing

Discovered

Unseen

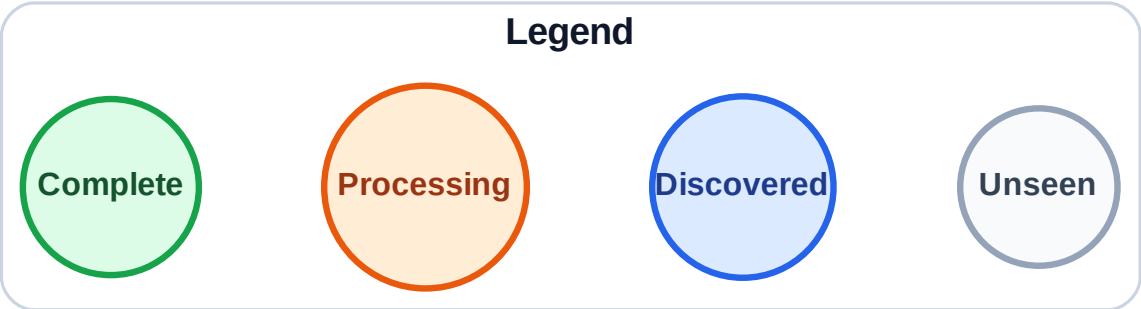
Queue: B → D → F → G

Visited: A, C, E, H



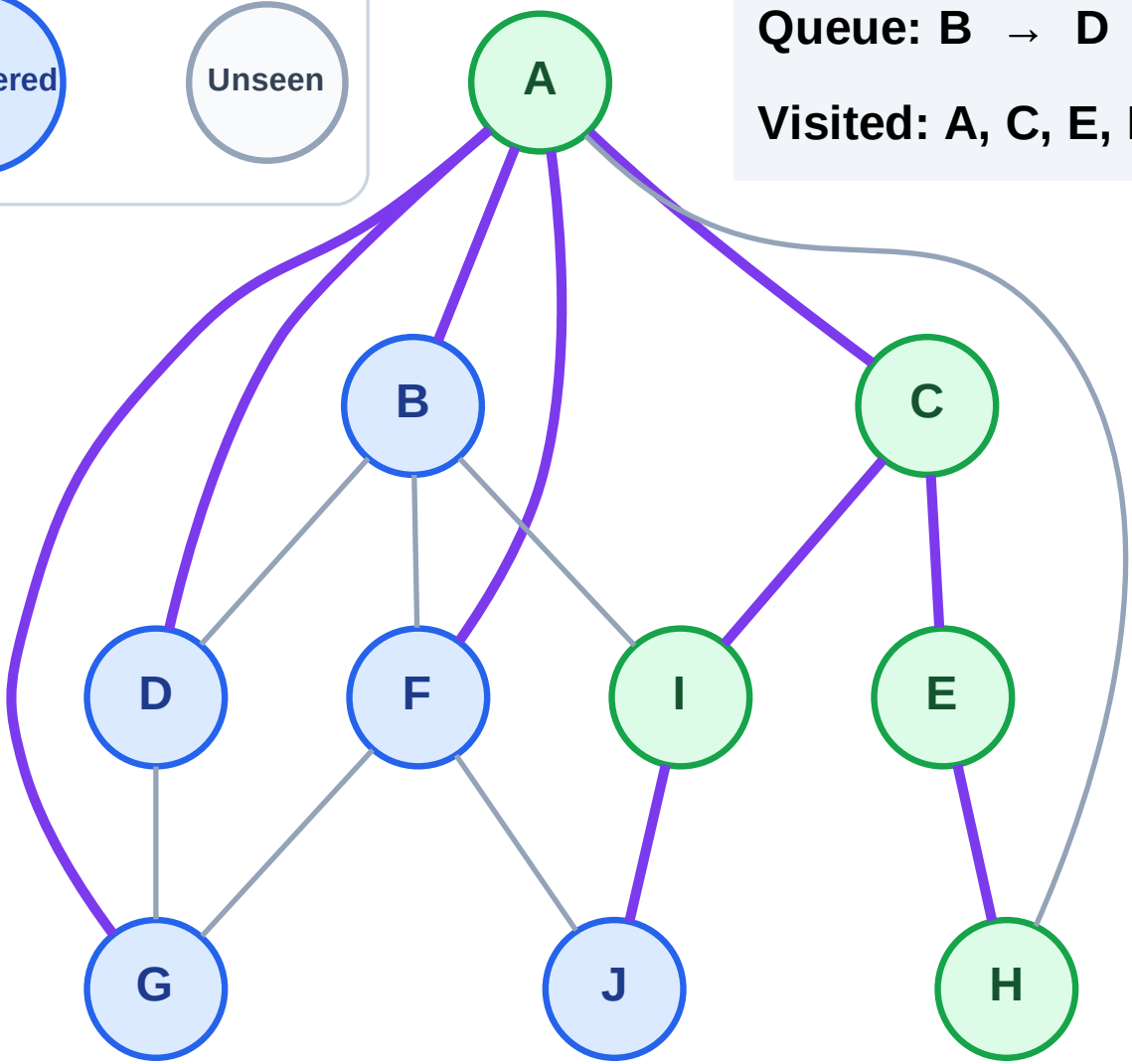
BFS Traversal

Step 11: Discover J from I



Queue: B → D → F → G → J

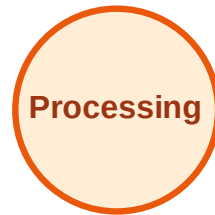
Visited: A, C, E, H, I



BFS Traversal

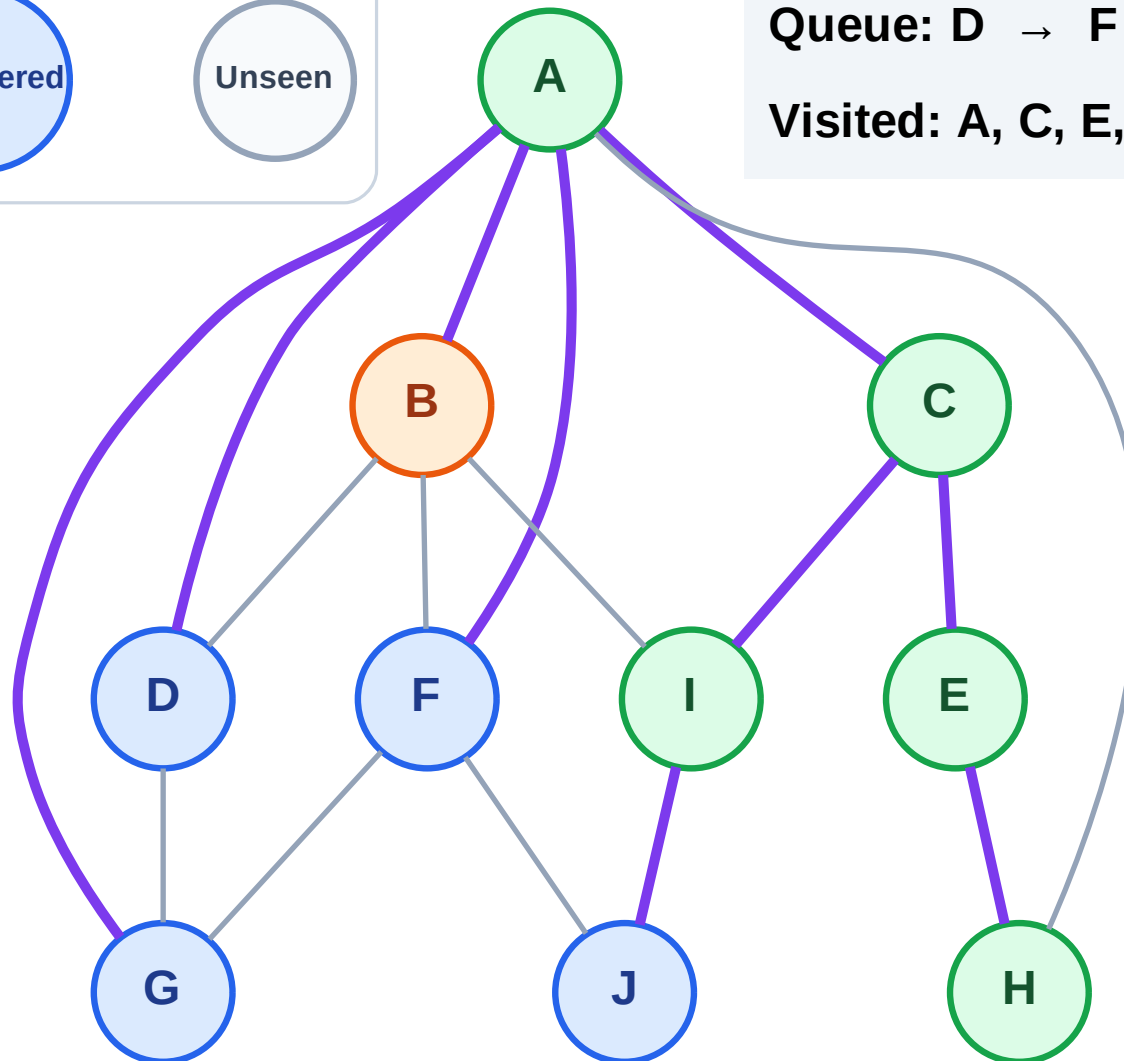
Step 12: Process node B

Legend



Queue: D → F → G → J

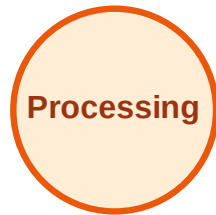
Visited: A, C, E, H, I



BFS Traversal

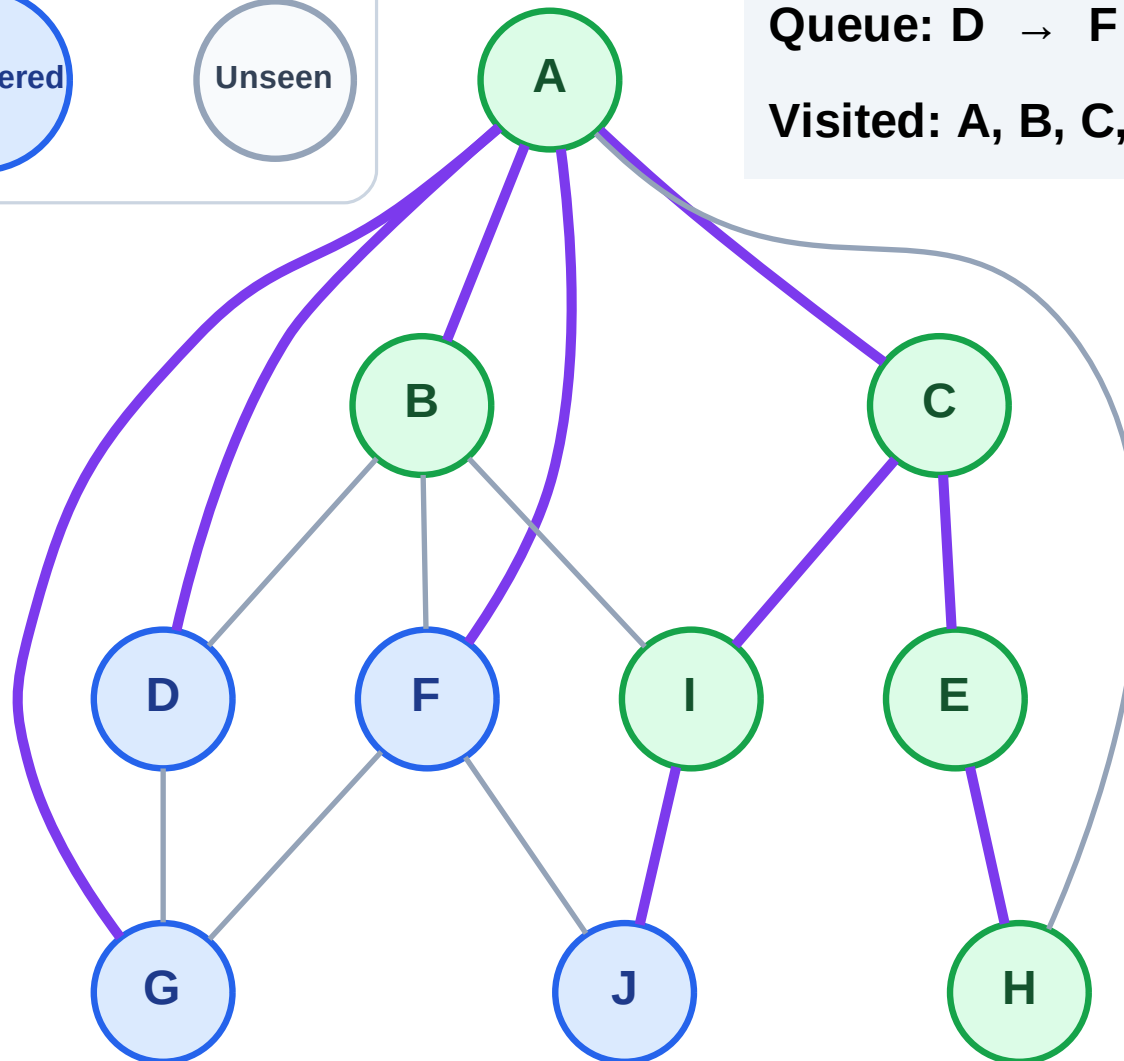
Step 13: No new neighbors from B

Legend



Queue: D → F → G → J

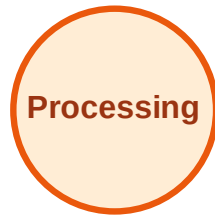
Visited: A, B, C, E, H, I



BFS Traversal

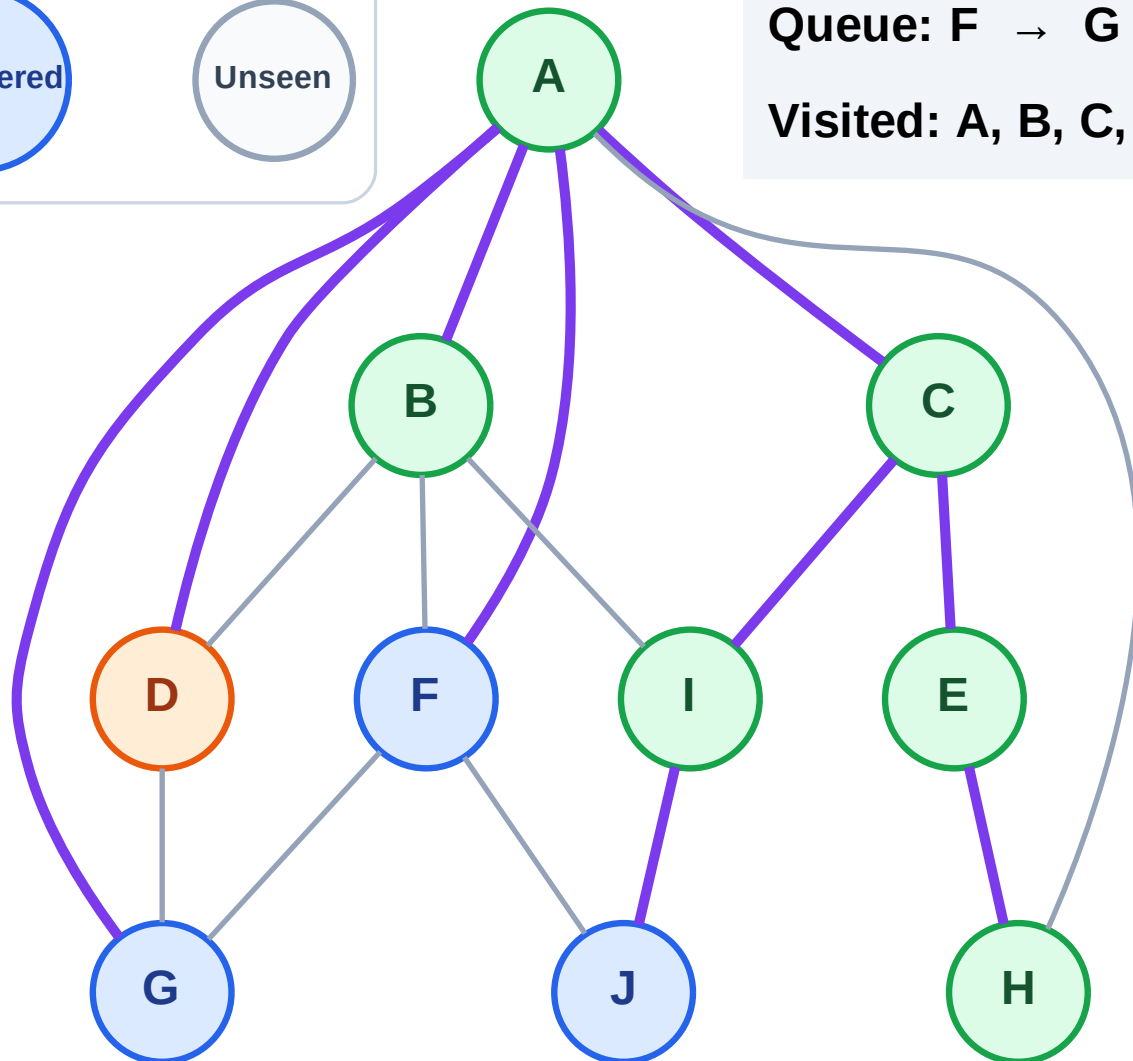
Step 14: Process node D

Legend



Queue: F → G → J

Visited: A, B, C, E, H, I



BFS Traversal

Step 15: No new neighbors from D

Legend

Complete

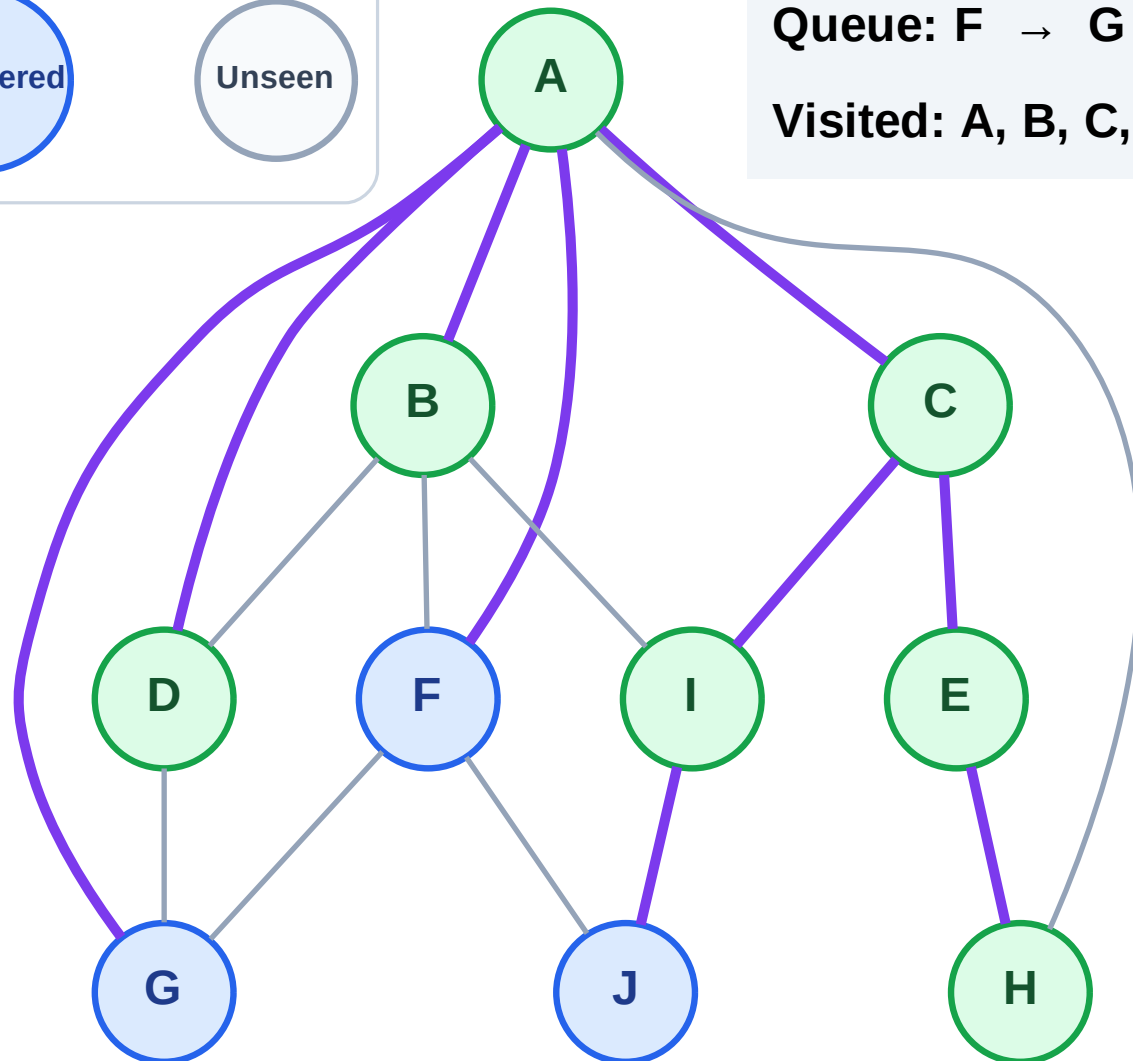
Processing

Discovered

Unseen

Queue: F → G → J

Visited: A, B, C, D, E, H, I



BFS Traversal

Step 16: Process node F

Legend

Complete

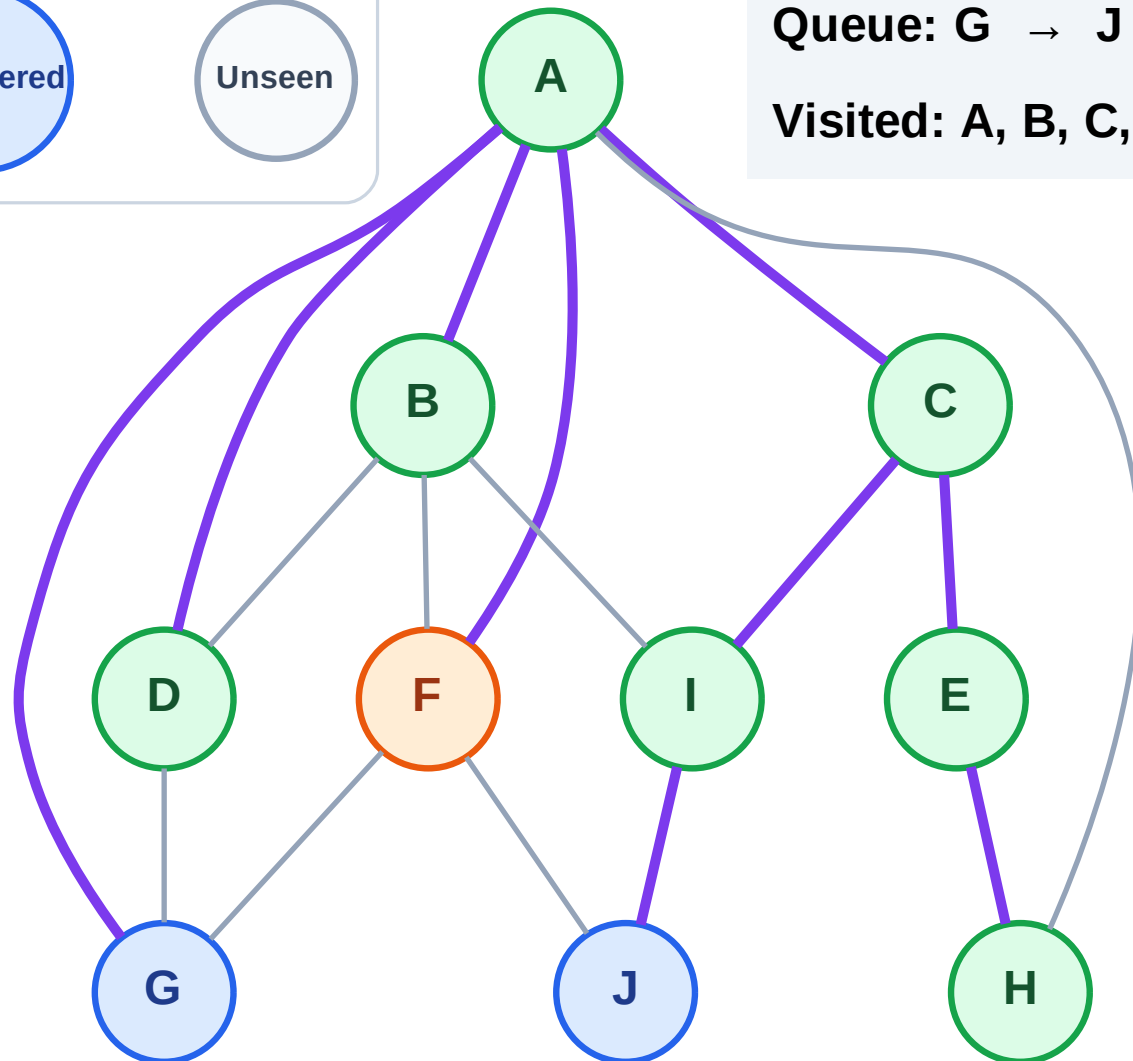
Processing

Discovered

Unseen

Queue: G → J

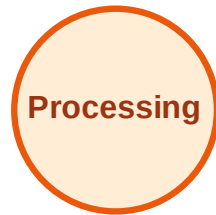
Visited: A, B, C, D, E, H, I



BFS Traversal

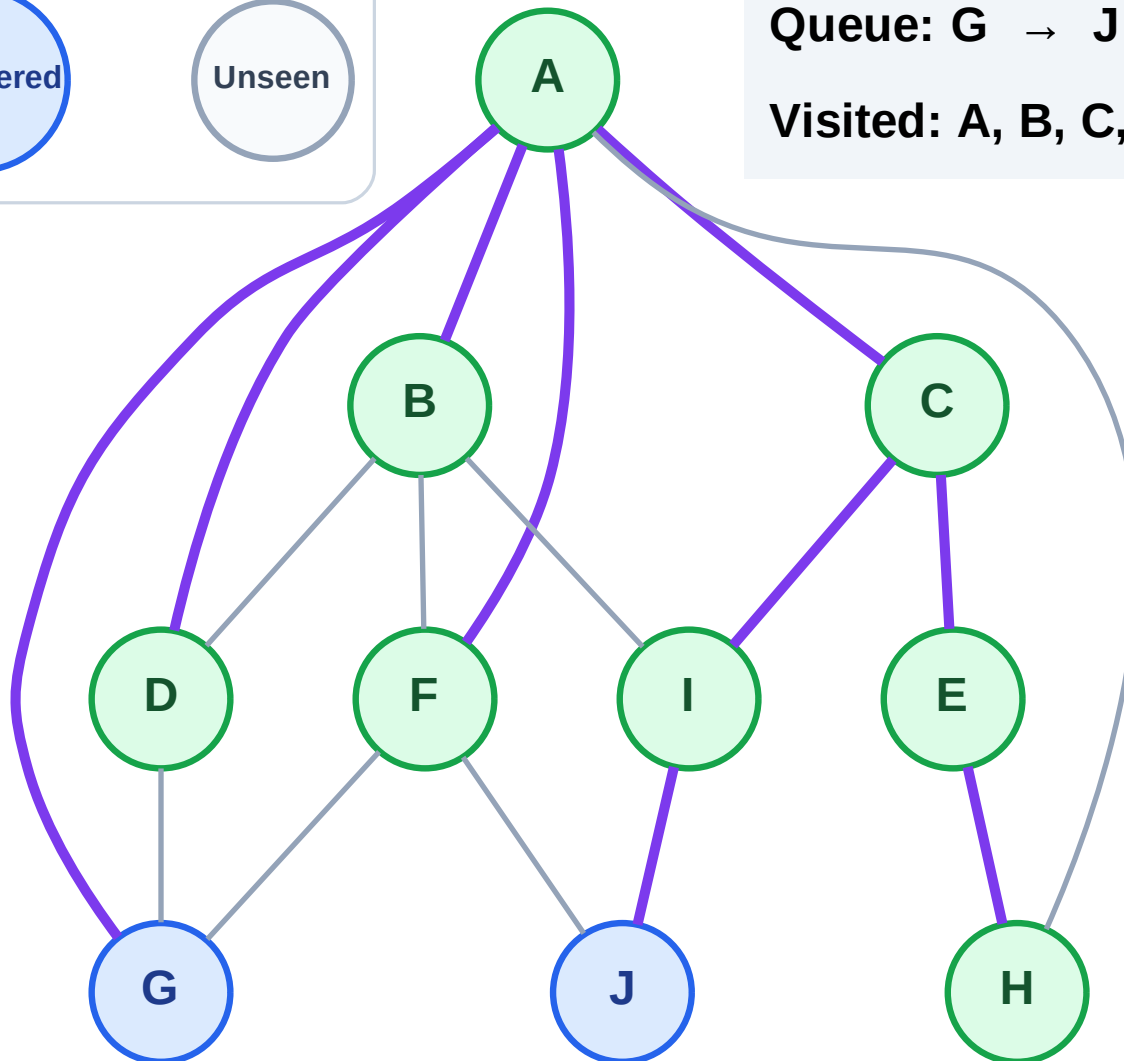
Step 17: No new neighbors from F

Legend



Queue: G → J

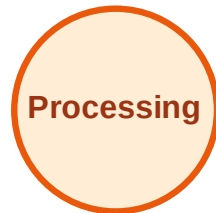
Visited: A, B, C, D, E, F, H, I



BFS Traversal

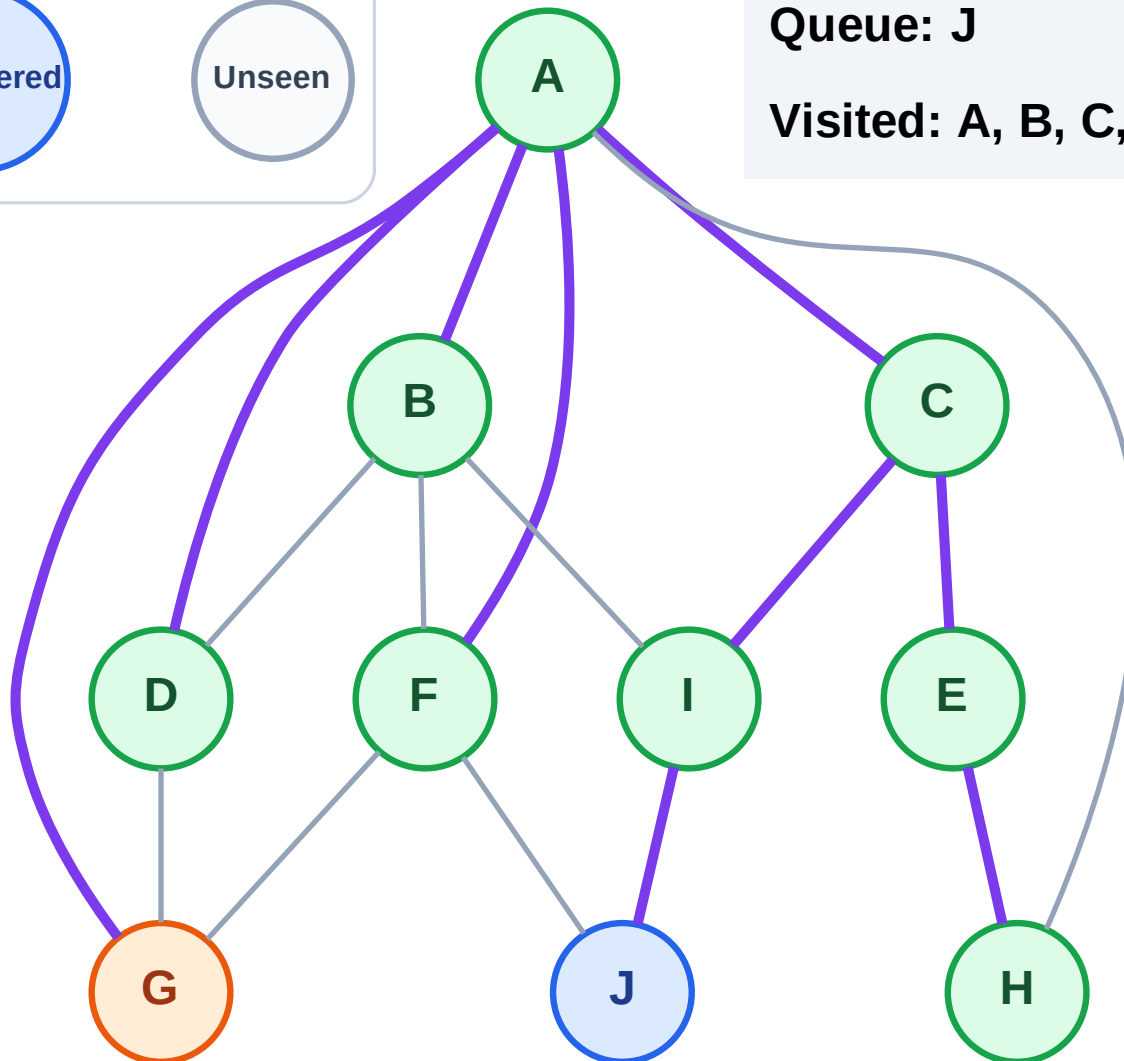
Step 18: Process node G

Legend



Queue: J

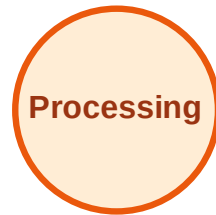
Visited: A, B, C, D, E, F, H, I



BFS Traversal

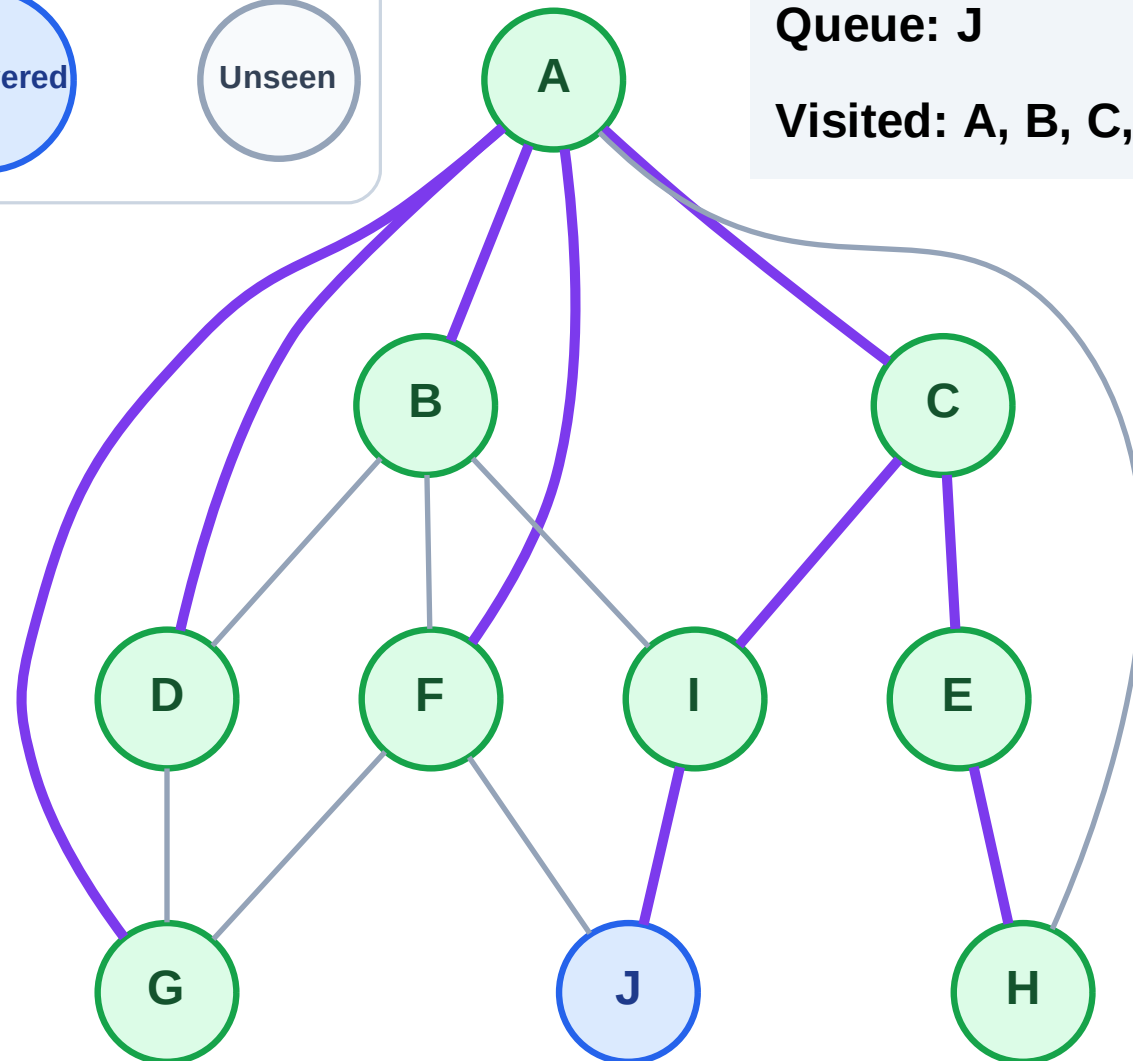
Step 19: No new neighbors from G

Legend



Queue: J

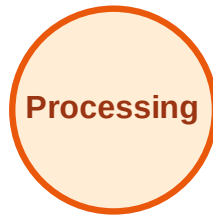
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

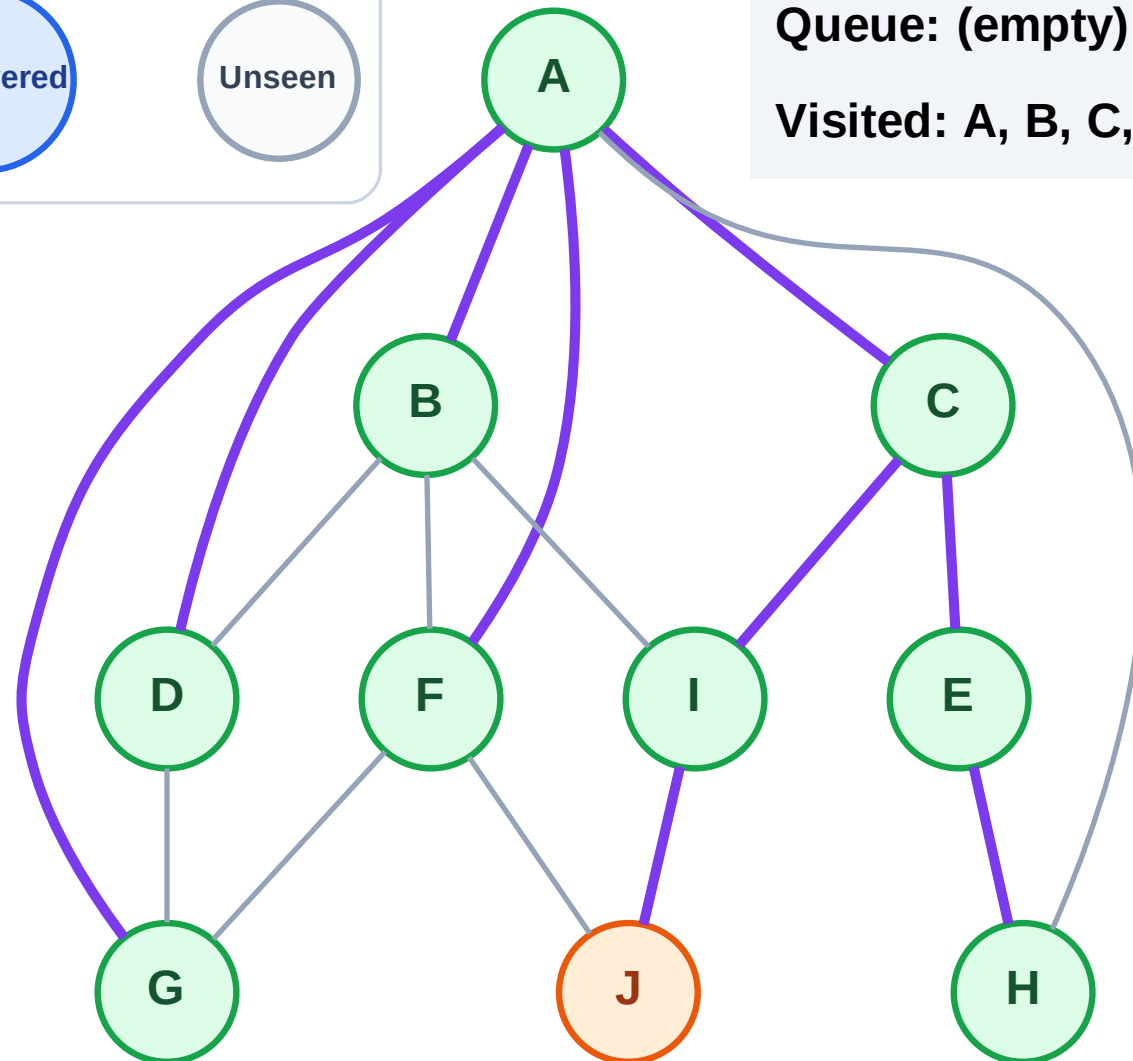
Step 20: Process node J

Legend



Queue: (empty)

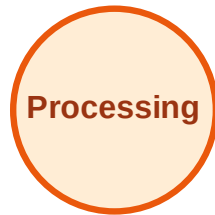
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

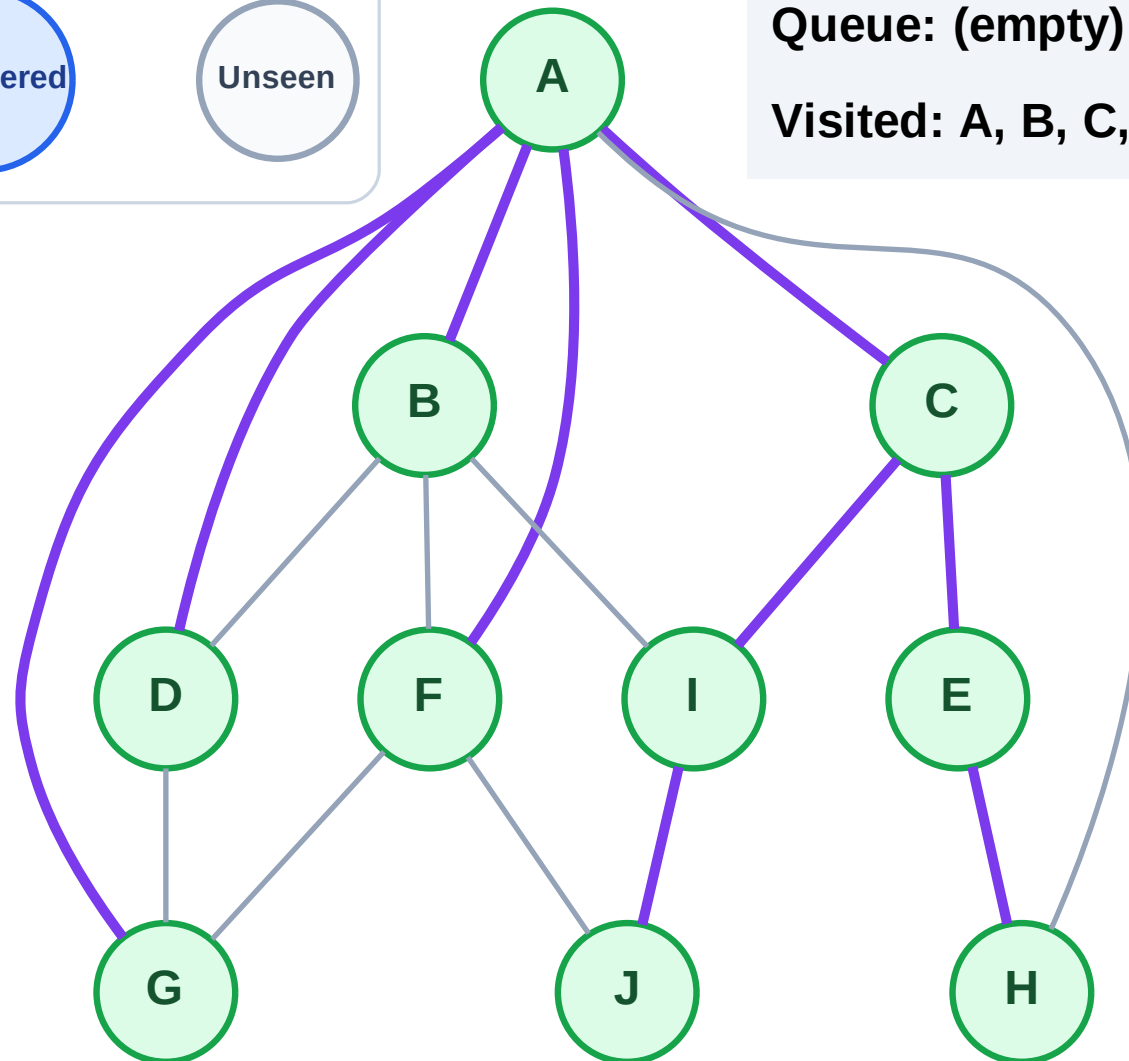
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

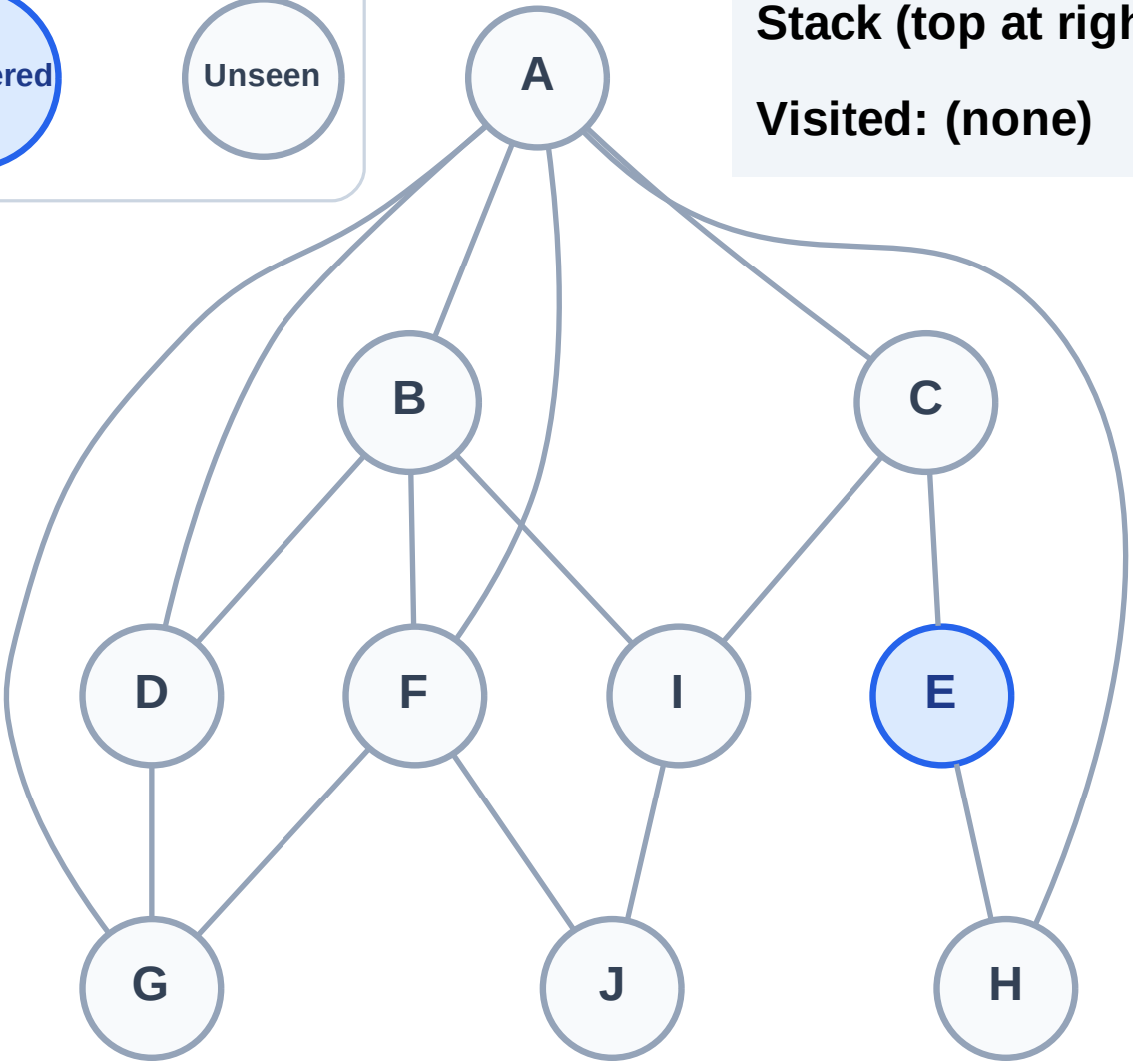
Step 1: Start at E

Legend



Stack (top at right): E

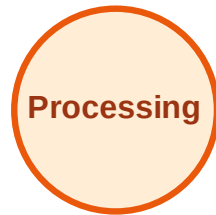
Visited: (none)



DFS Traversal

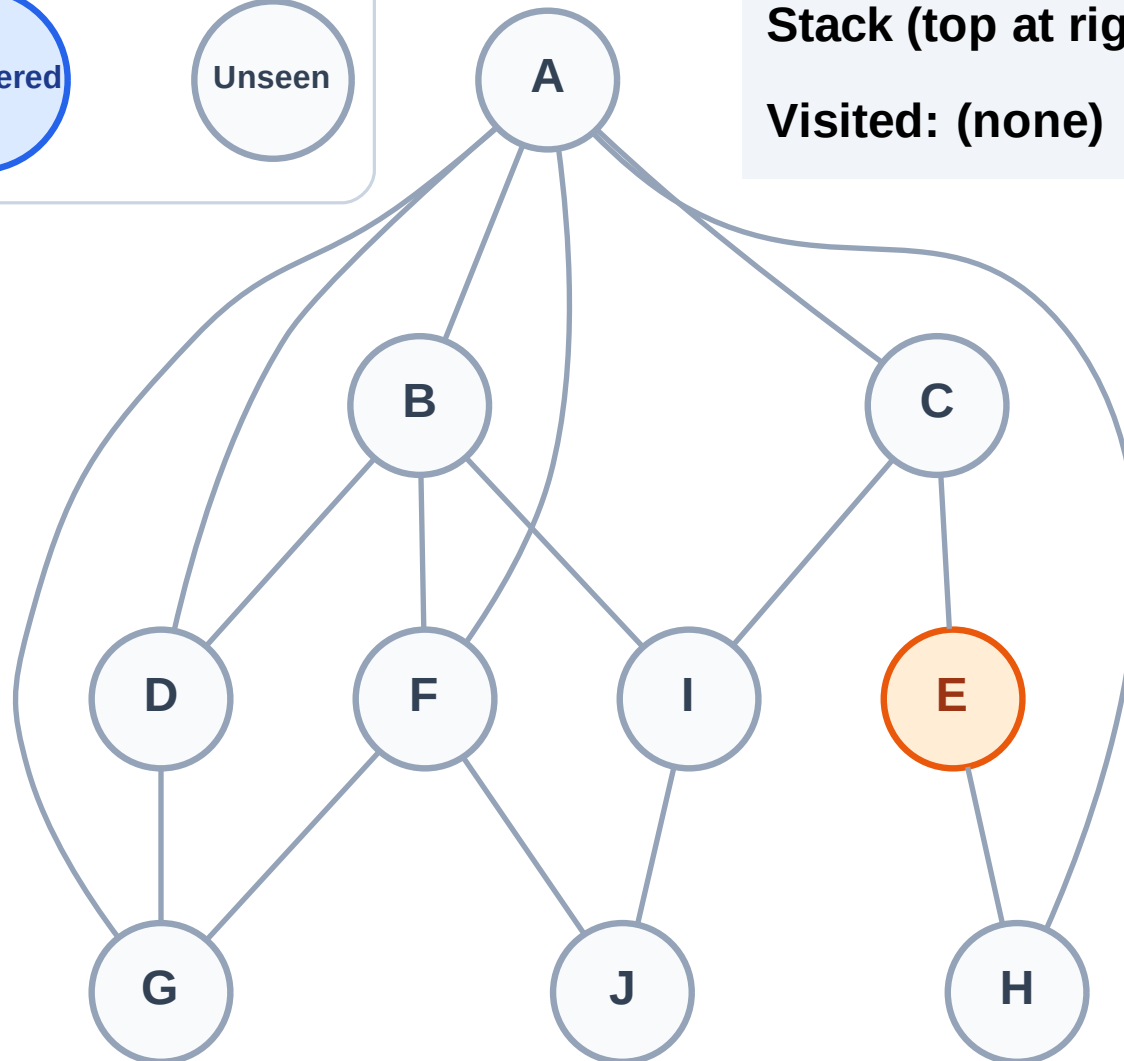
Step 2: Process node E

Legend



Stack (top at right): (empty)

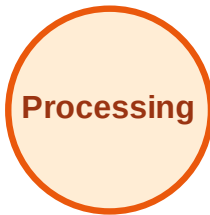
Visited: (none)



DFS Traversal

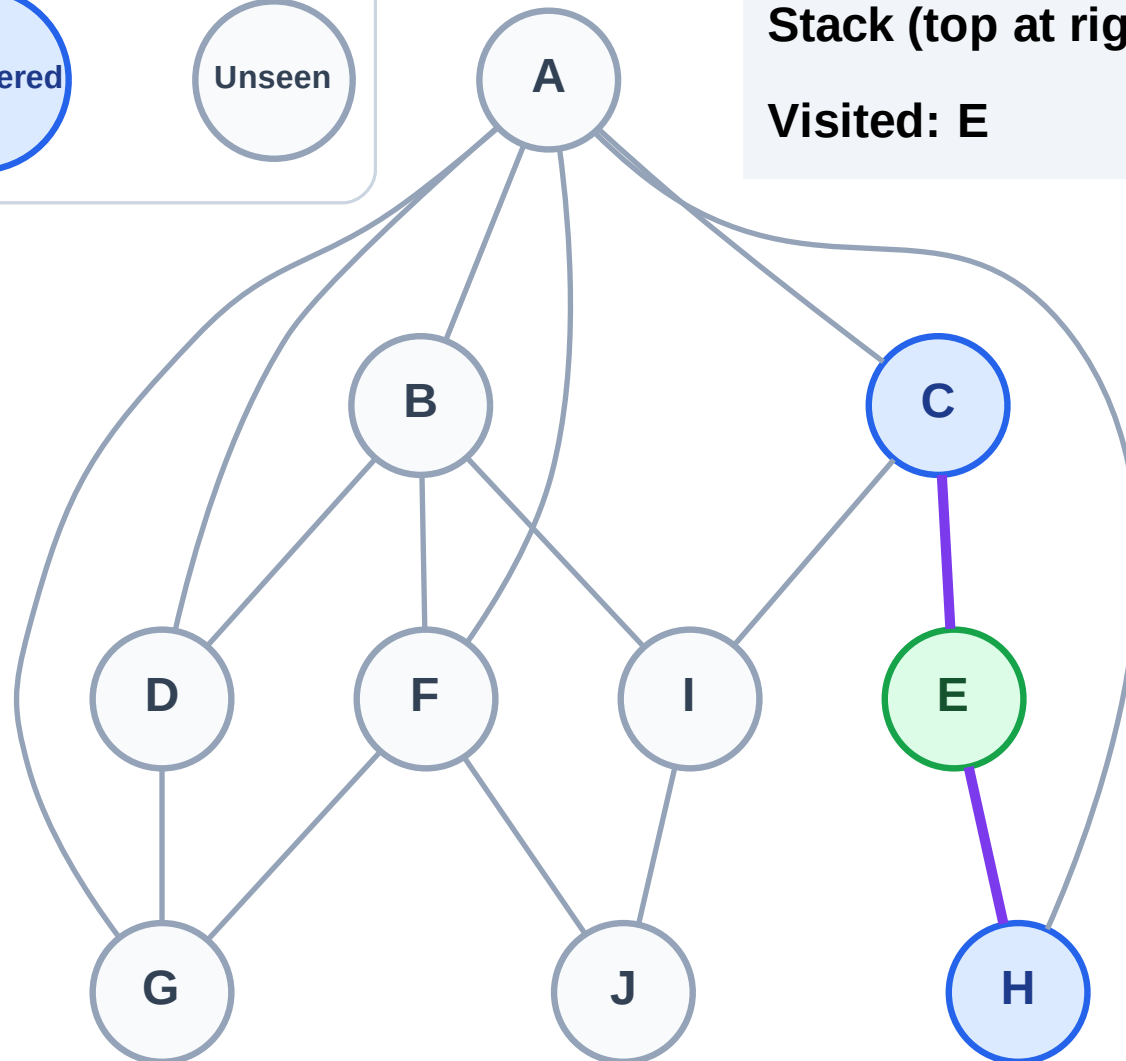
Step 3: Discover H, C from E

Legend



Stack (top at right): H | C

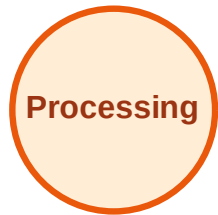
Visited: E



DFS Traversal

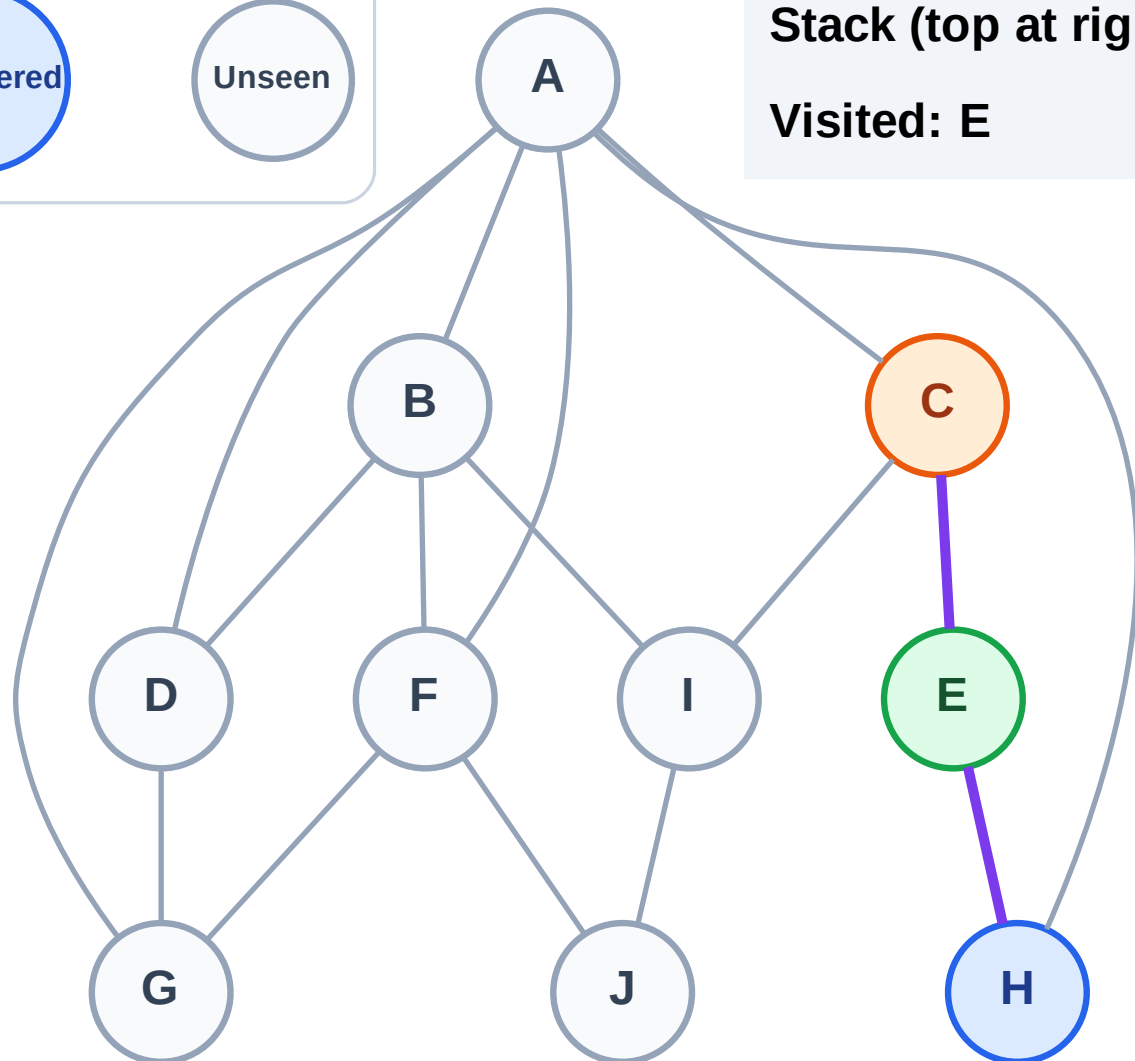
Step 4: Process node C

Legend



Stack (top at right): H

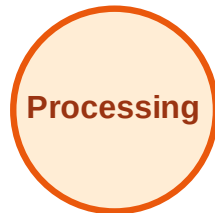
Visited: E



DFS Traversal

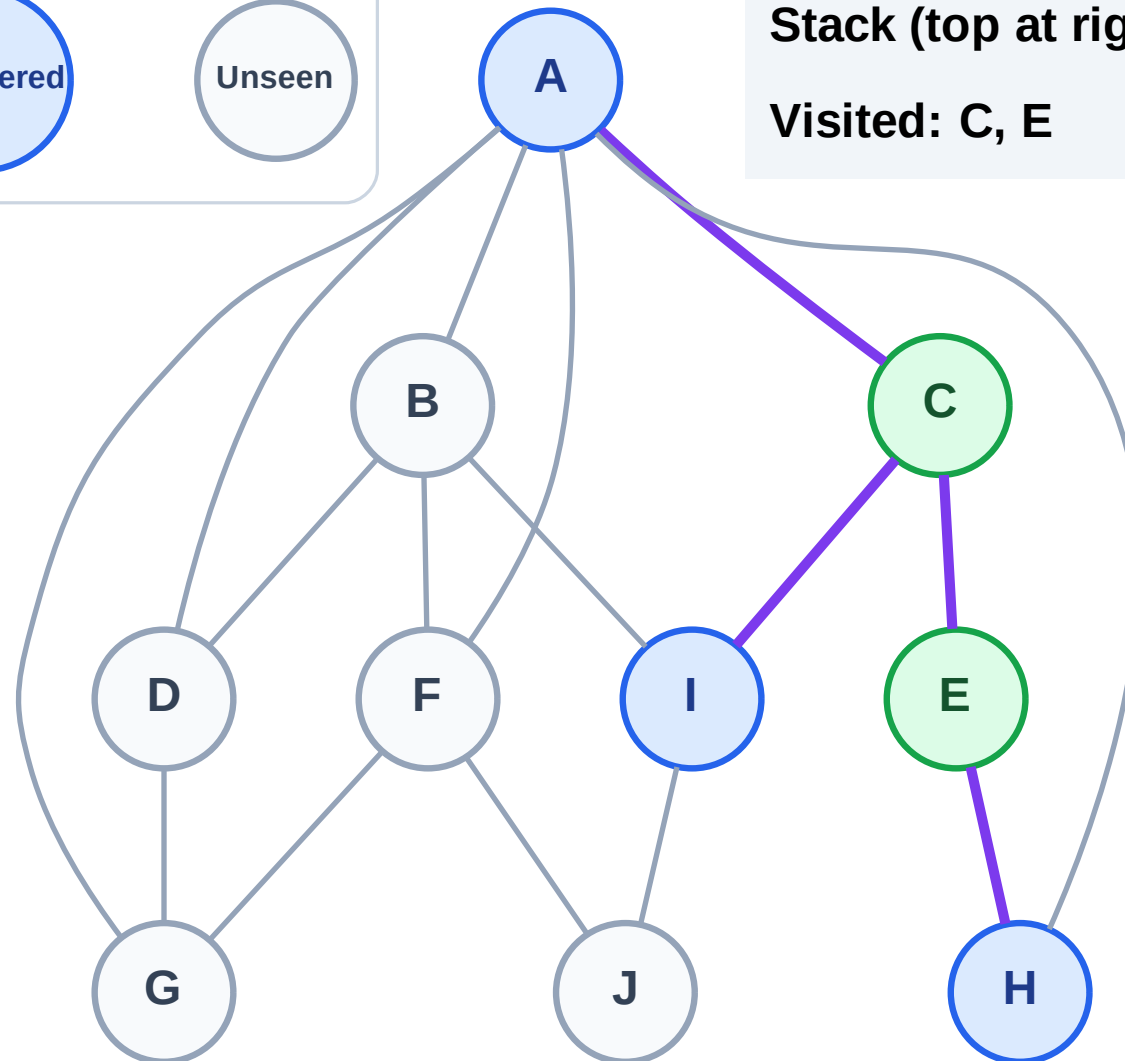
Step 5: Discover I, A from C

Legend



Stack (top at right): H | I | A

Visited: C, E



DFS Traversal

Step 6: Process node A

Legend

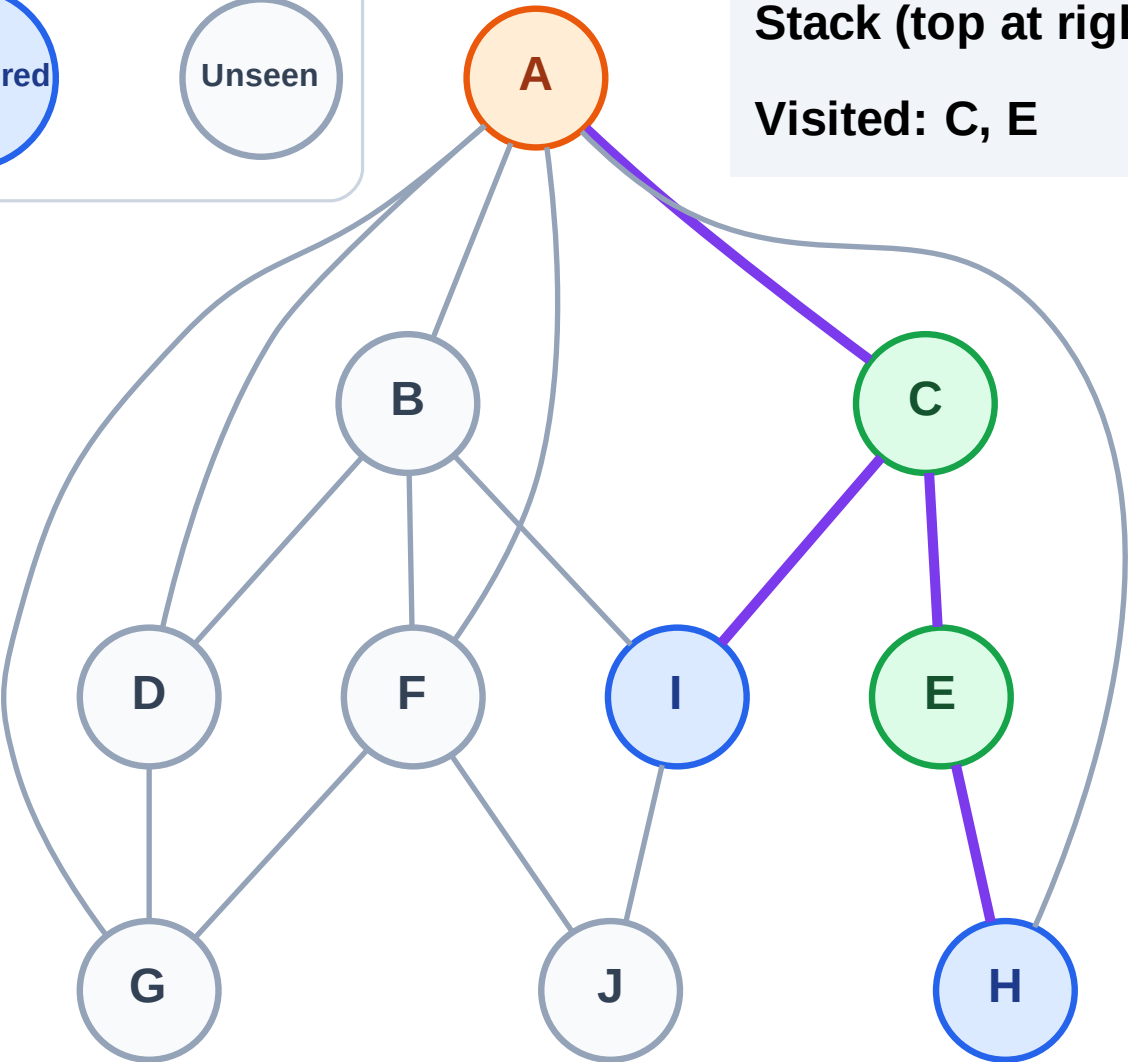
Complete

Processing

Discovered

Unseen

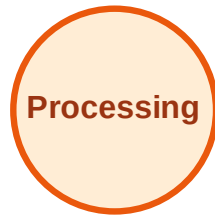
Stack (top at right): H | I
Visited: C, E



DFS Traversal

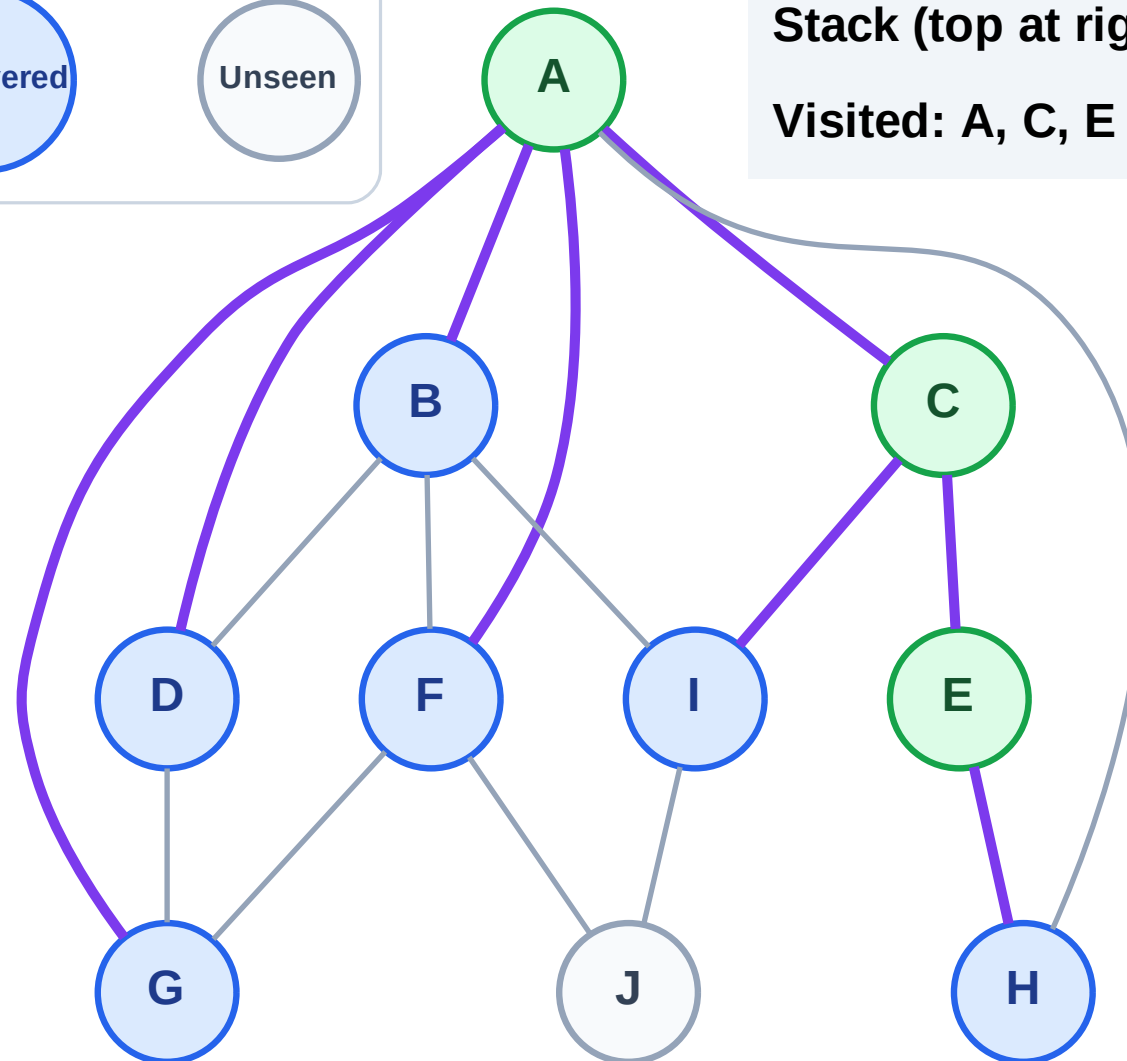
Step 7: Discover G, F, D, B from A

Legend



Stack (top at right): H | I | G | F | D | B

Visited: A, C, E



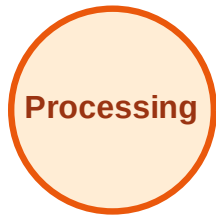
DFS Traversal

Step 8: Process node B

Legend



Complete



Processing



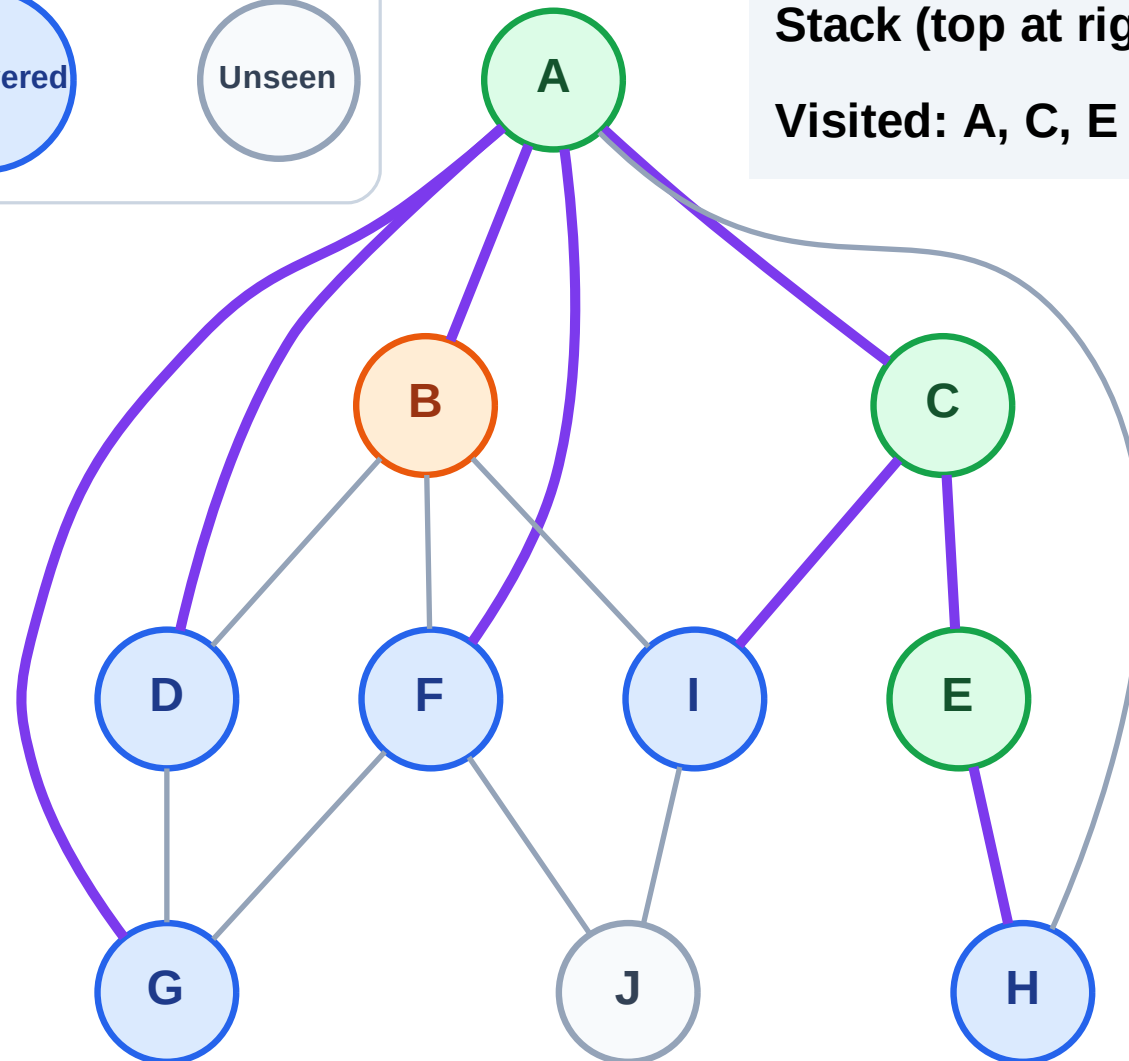
Discovered



Unseen

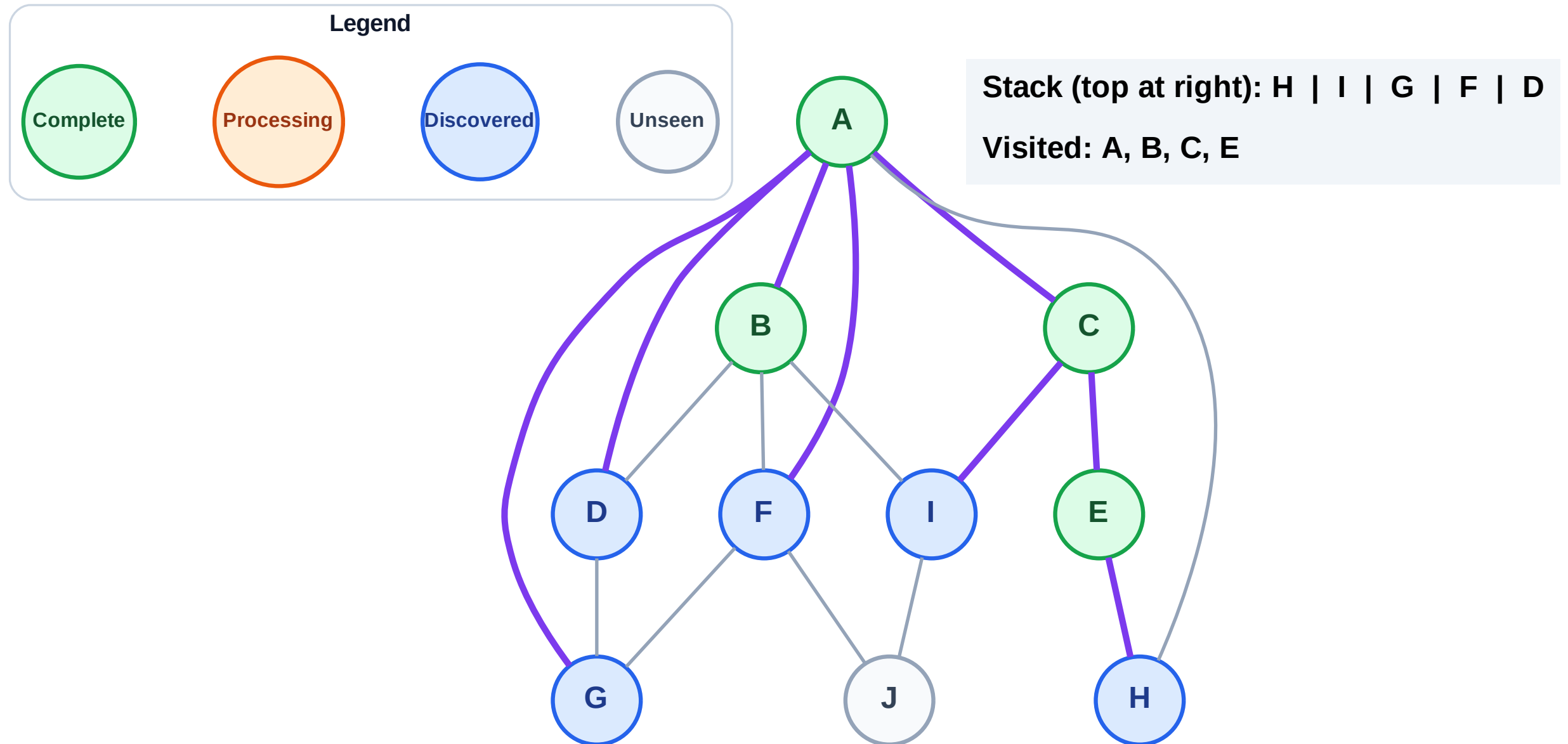
Stack (top at right): H | I | G | F | D

Visited: A, C, E



Step 9: No new neighbors from B

Step 9: No new neighbors from B



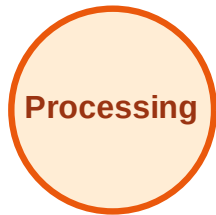
DFS Traversal

Step 10: Process node D

Legend



Complete



Processing



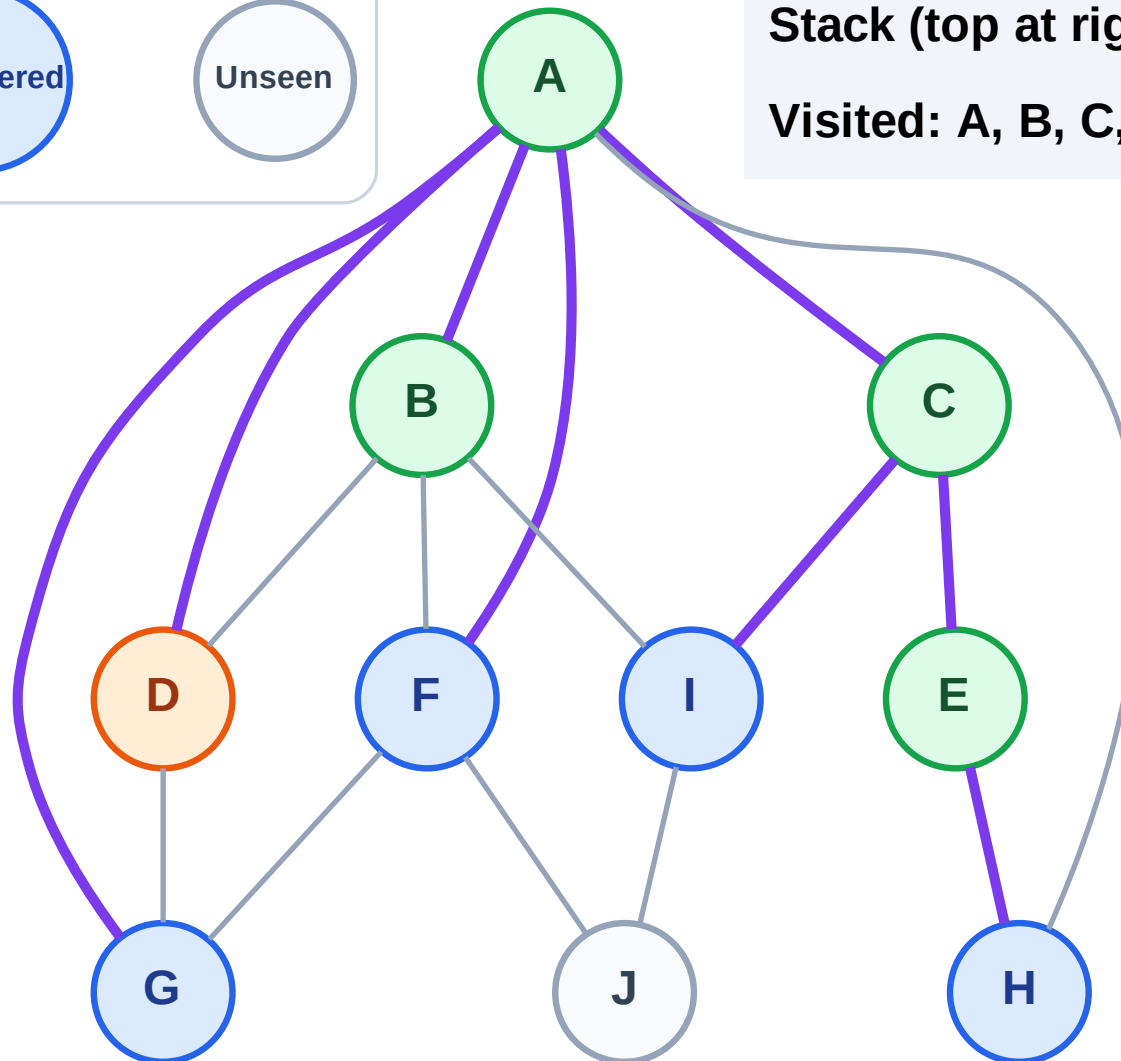
Discovered



Unseen

Stack (top at right): H | I | G | F

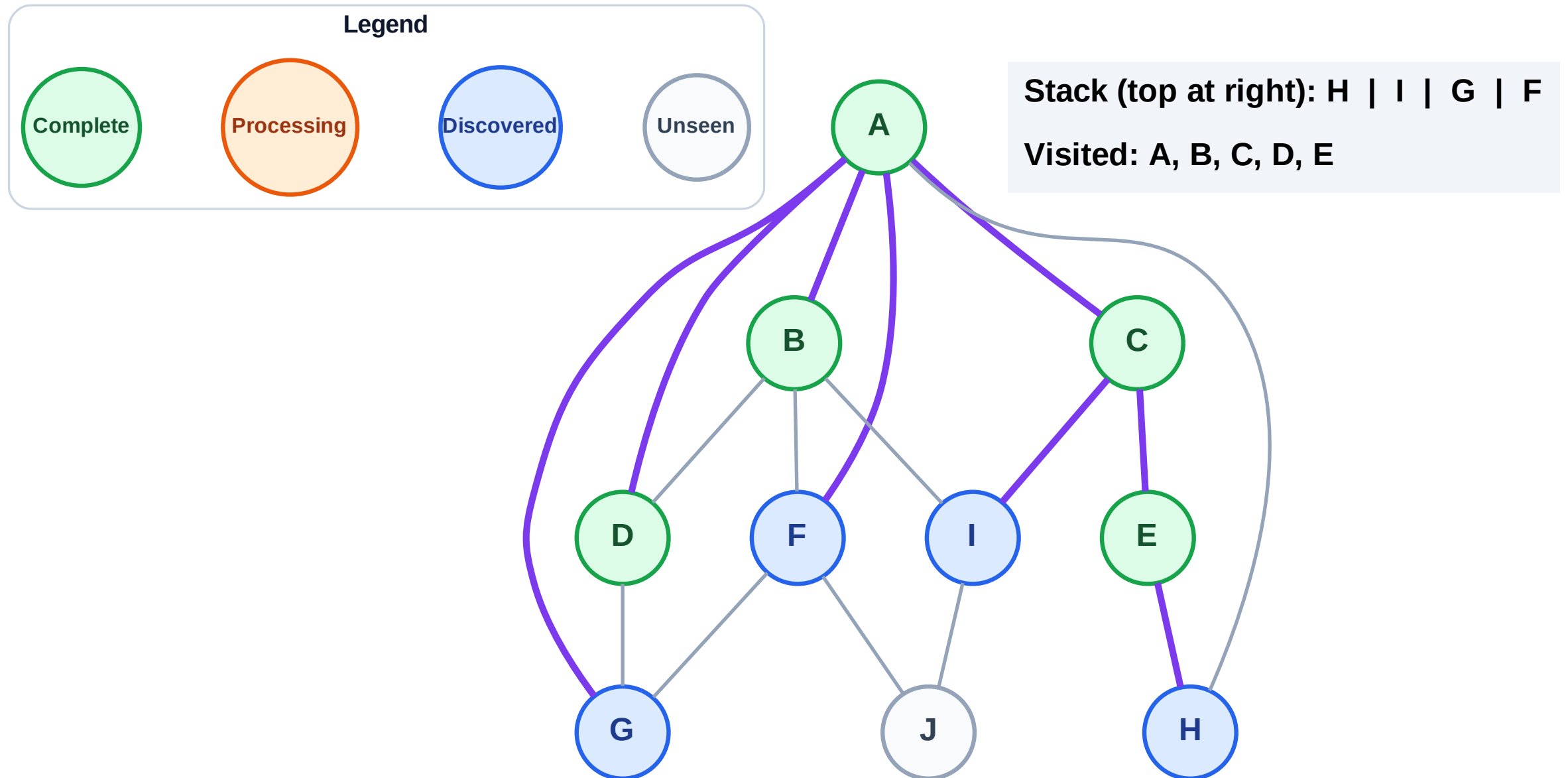
Visited: A, B, C, E



DFS Traversal

Step 11: No new neighbors from D

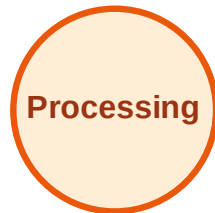
Step 11: No new neighbors from D



DFS Traversal

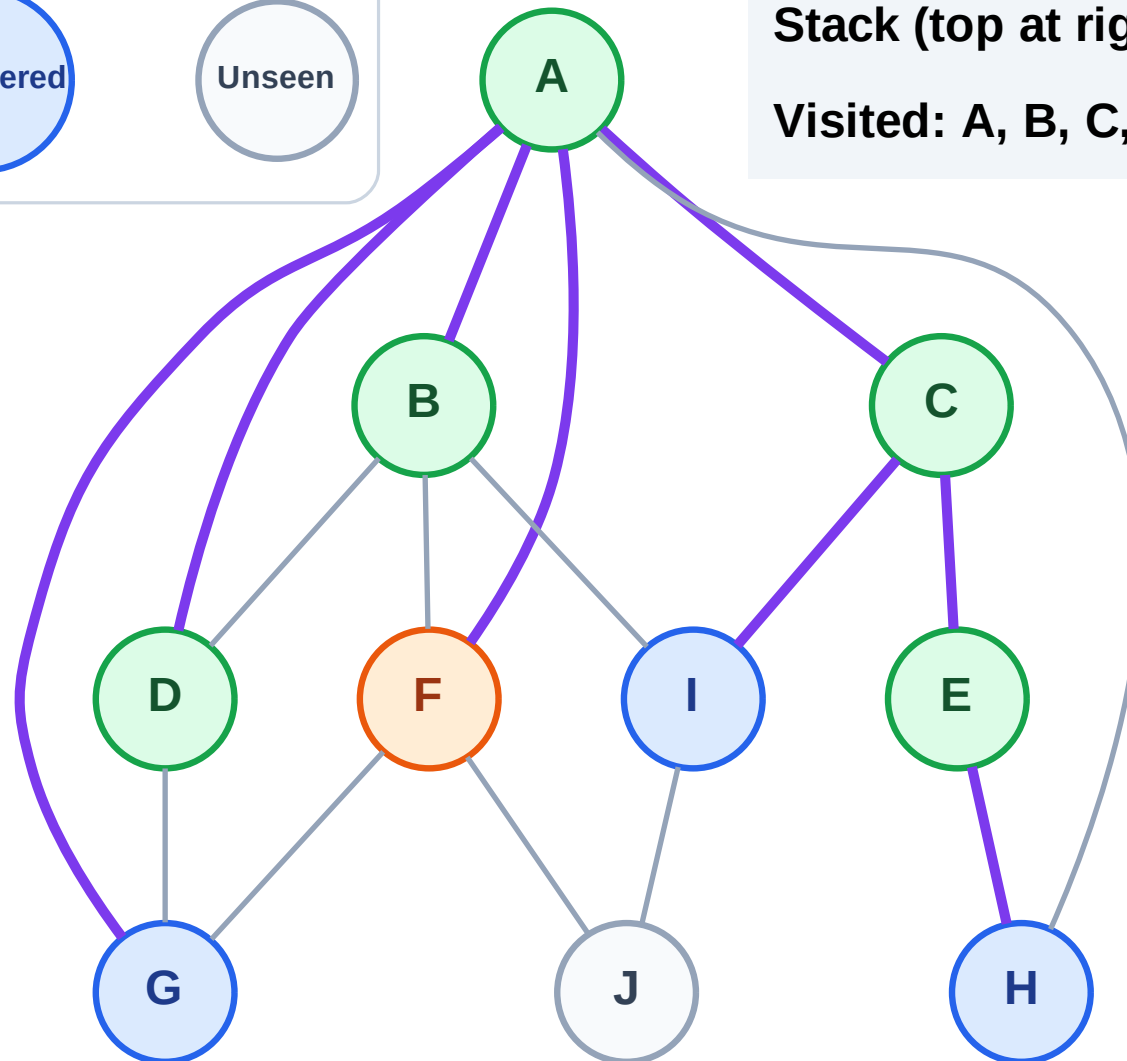
Step 12: Process node F

Legend



Stack (top at right): H | I | G

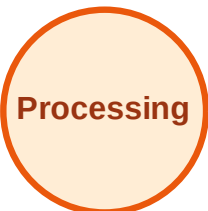
Visited: A, B, C, D, E



DFS Traversal

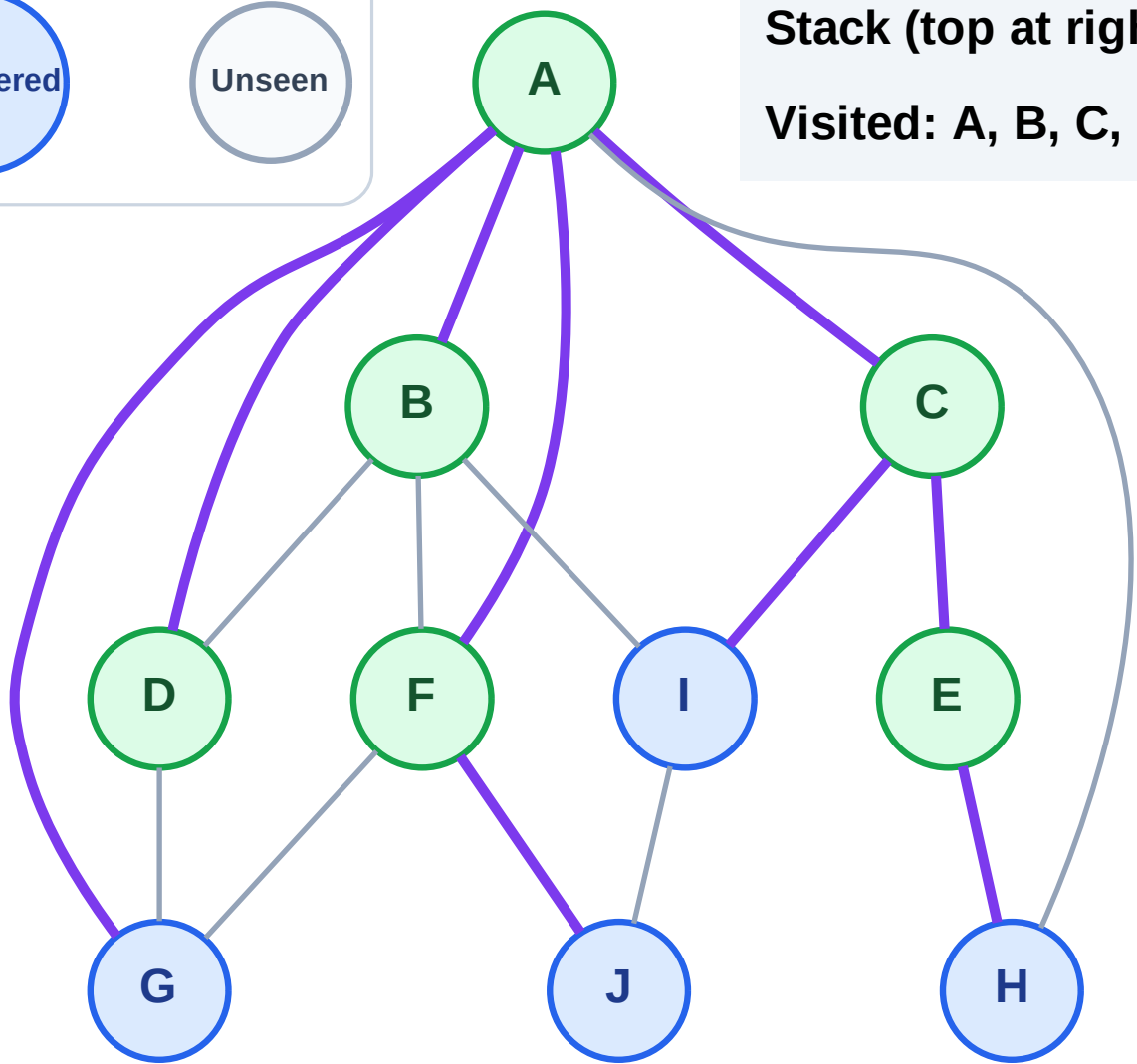
Step 13: Discover J from F

Legend



Stack (top at right): H | I | G | J

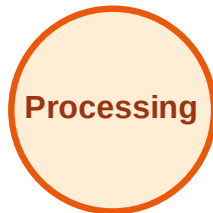
Visited: A, B, C, D, E, F



DFS Traversal

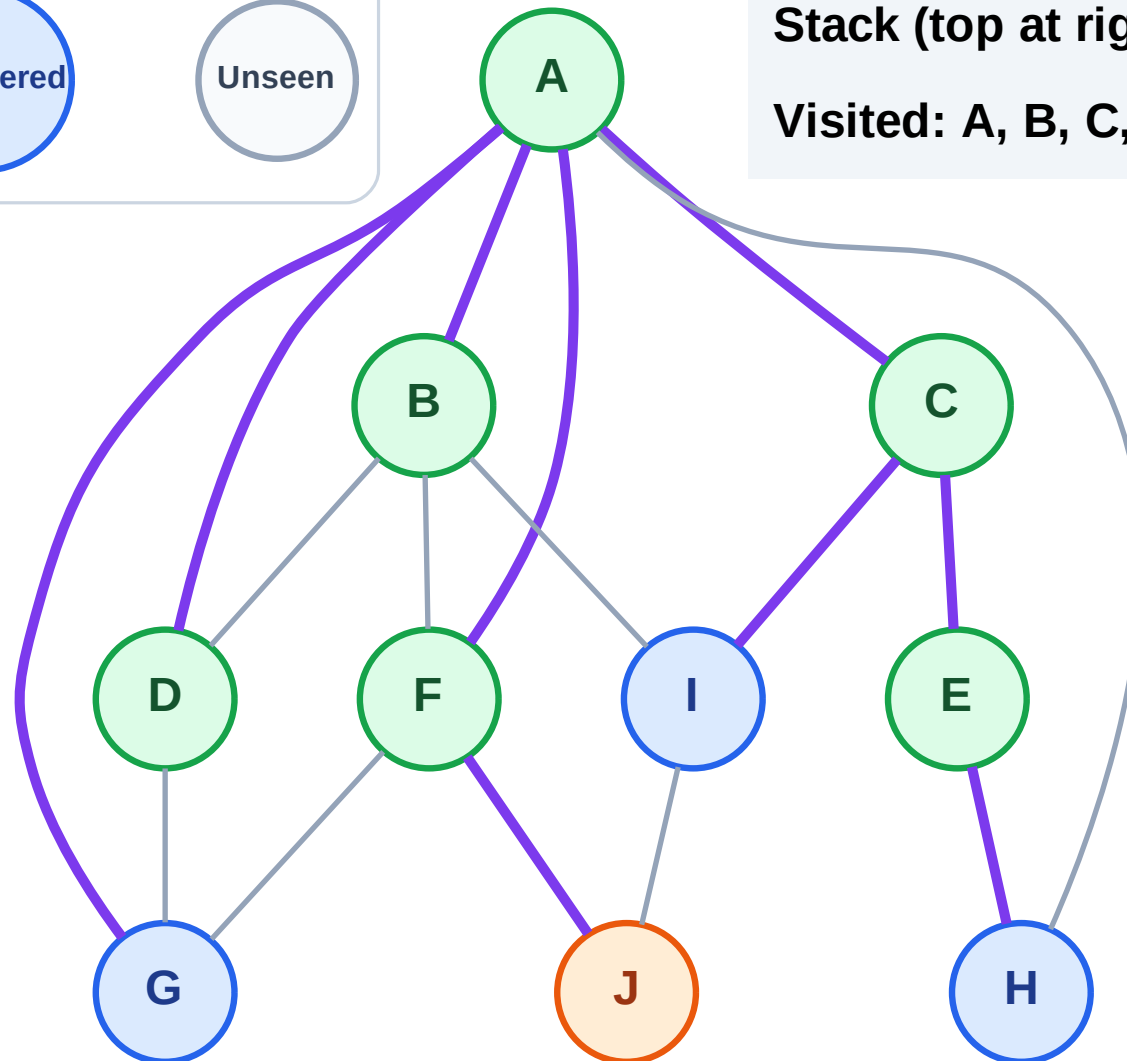
Step 14: Process node J

Legend



Stack (top at right): H | I | G

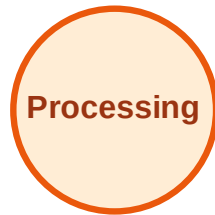
Visited: A, B, C, D, E, F



DFS Traversal

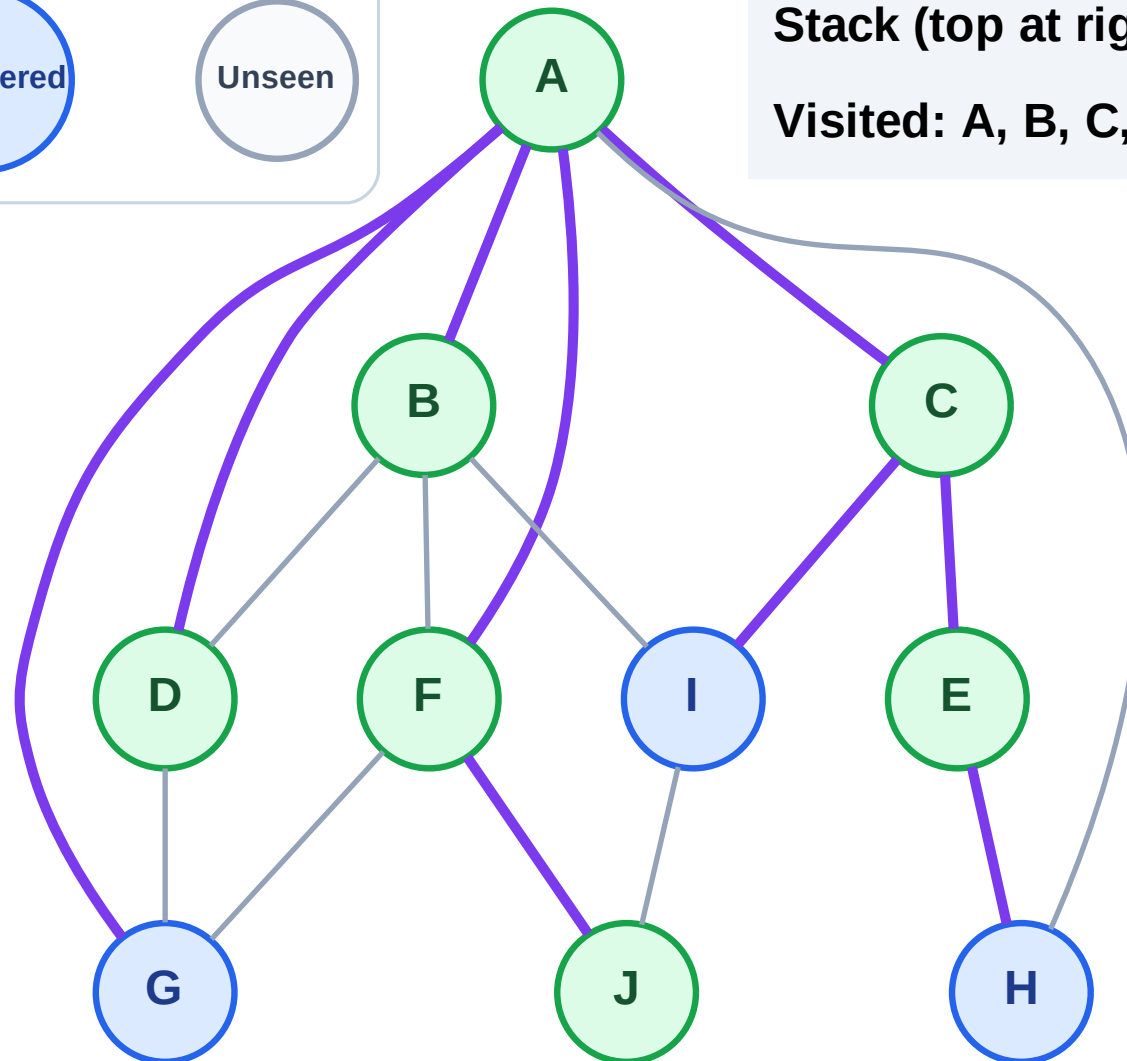
Step 15: No new neighbors from J

Legend



Stack (top at right): H | I | G

Visited: A, B, C, D, E, F, J



DFS Traversal

Step 16: Process node G

Legend

Complete

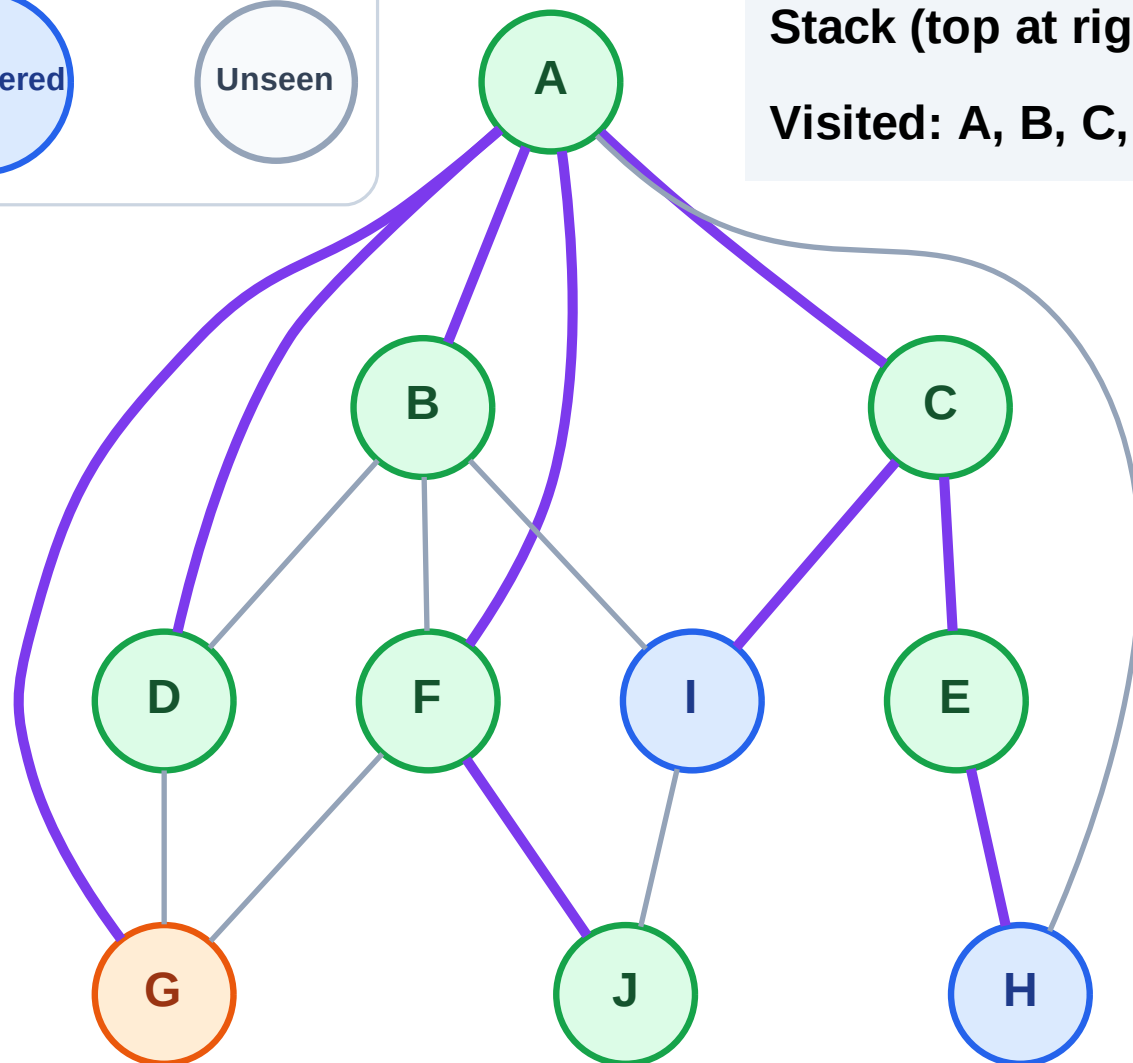
Processing

Discovered

Unseen

Stack (top at right): H | I

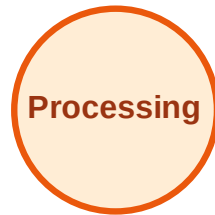
Visited: A, B, C, D, E, F, J



DFS Traversal

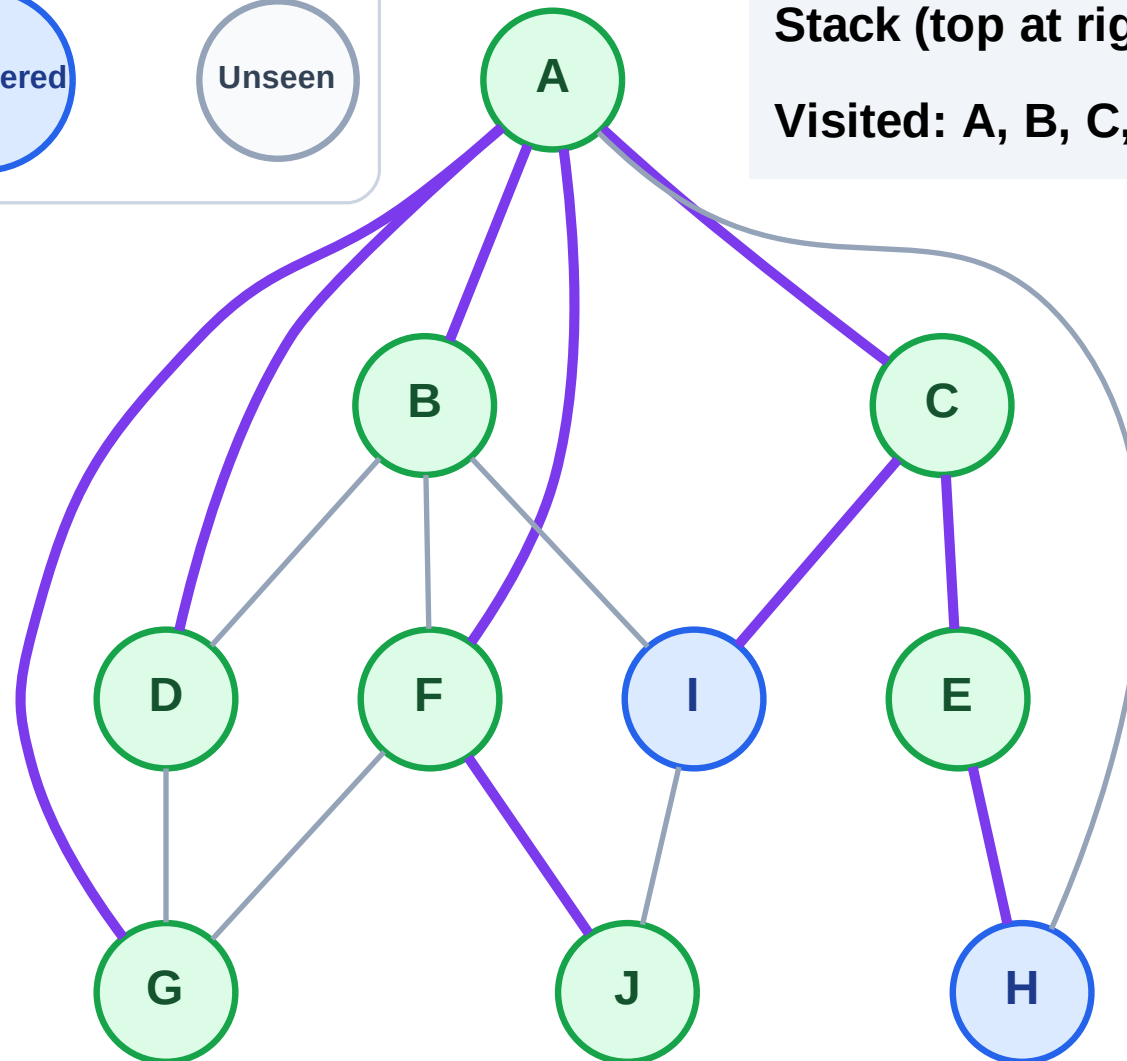
Step 17: No new neighbors from G

Legend



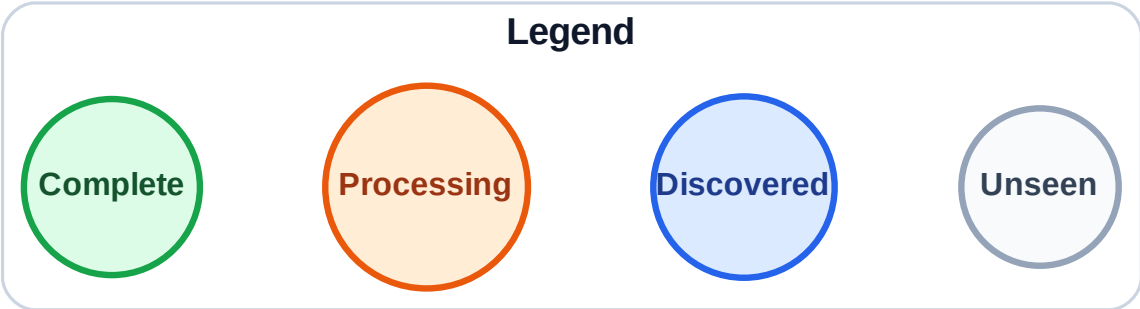
Stack (top at right): H | I

Visited: A, B, C, D, E, F, G, J

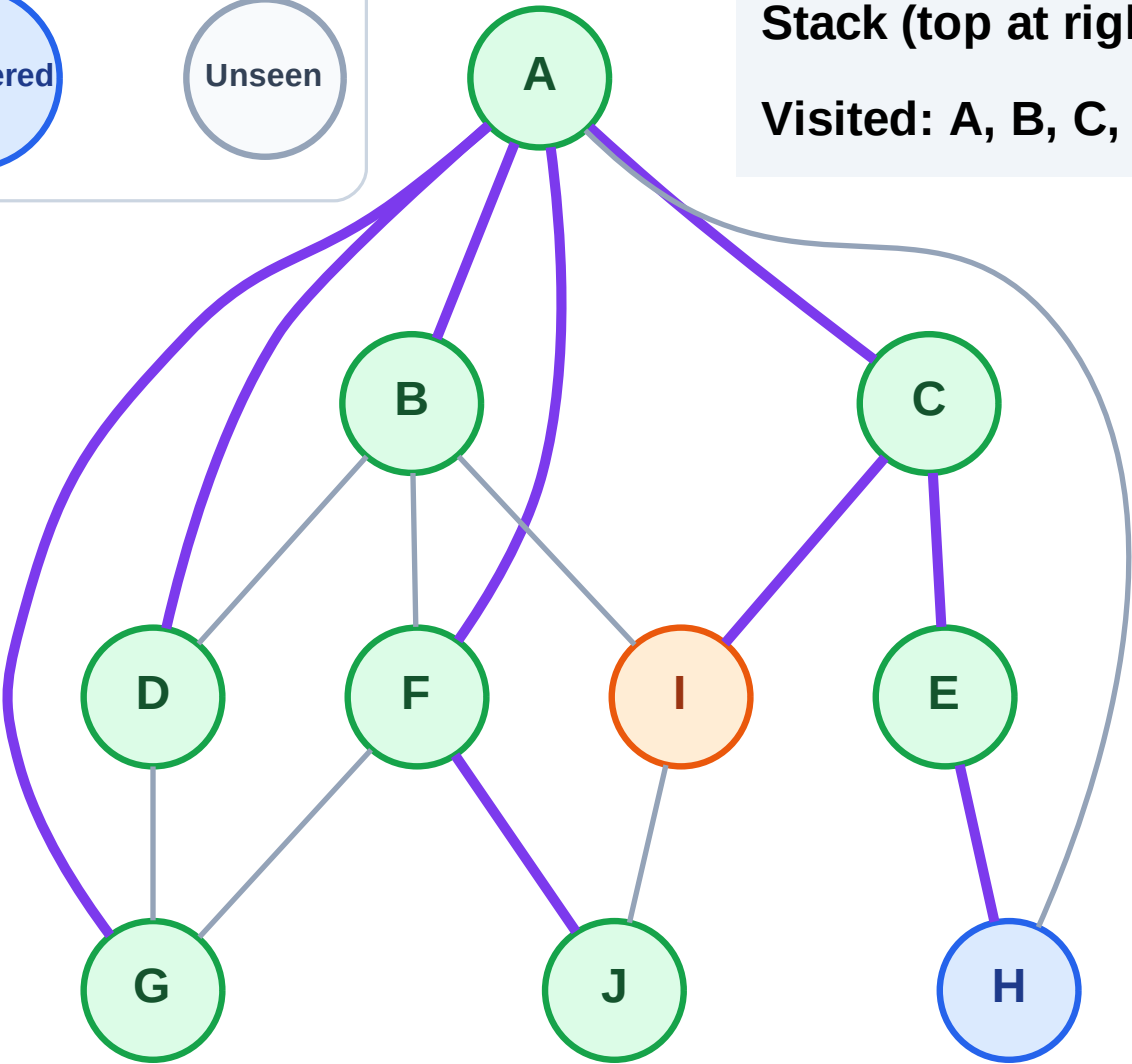


DFS Traversal

Step 18: Process node I



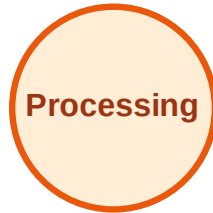
Stack (top at right): H
Visited: A, B, C, D, E, F, G, J



DFS Traversal

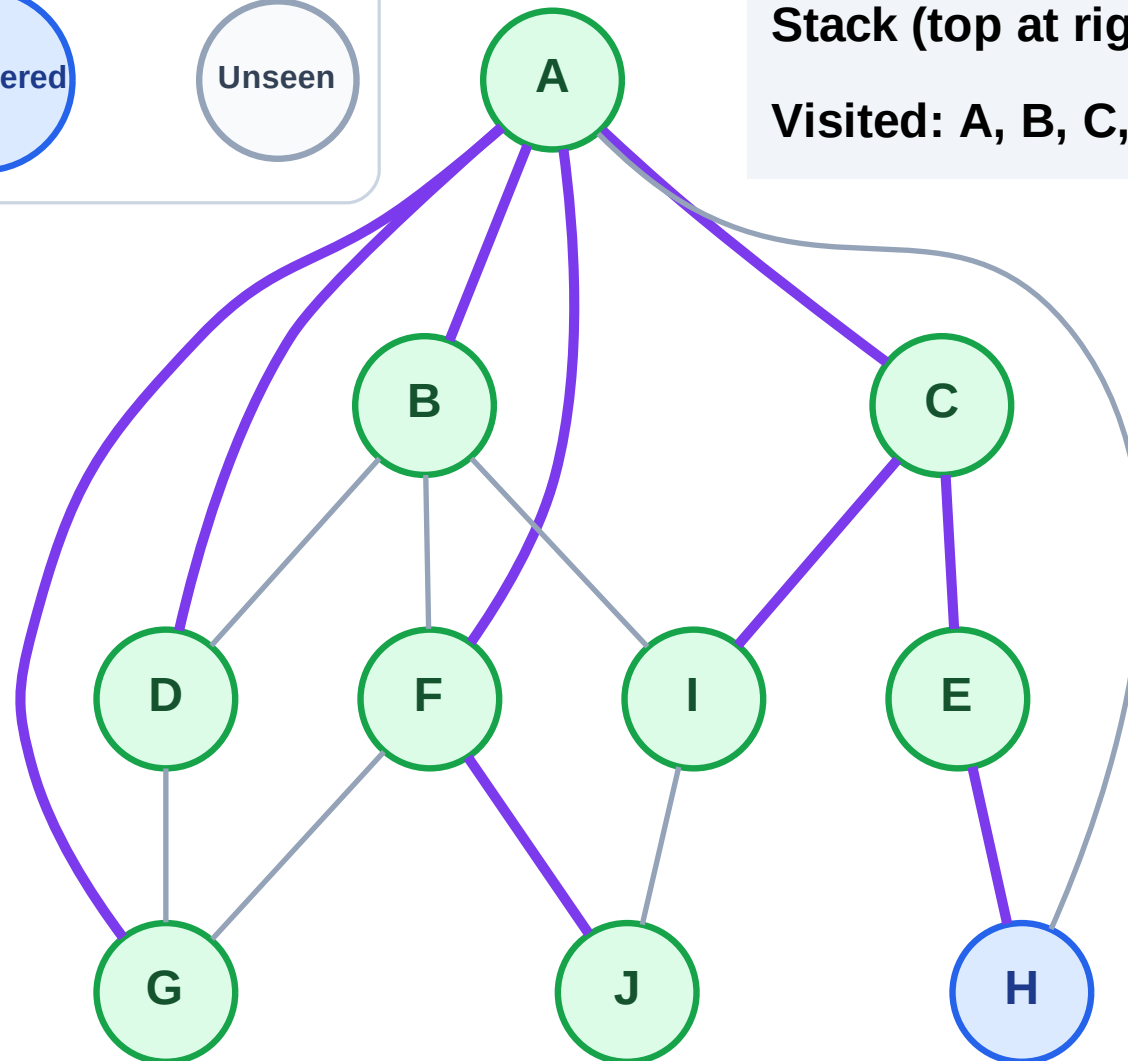
Step 19: No new neighbors from I

Legend



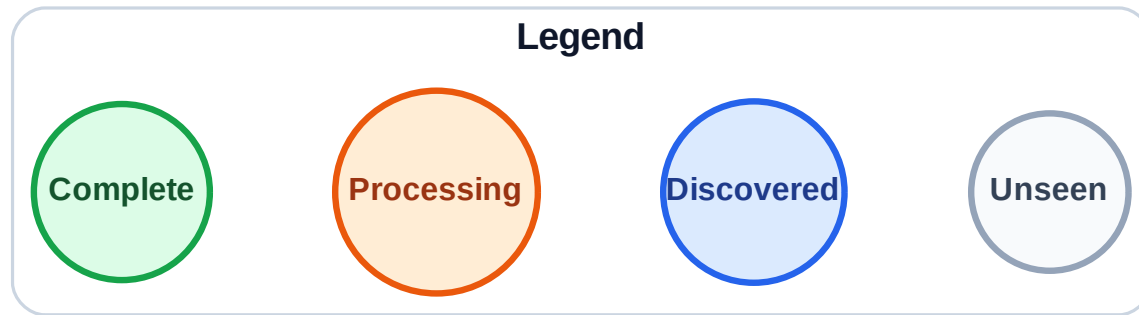
Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



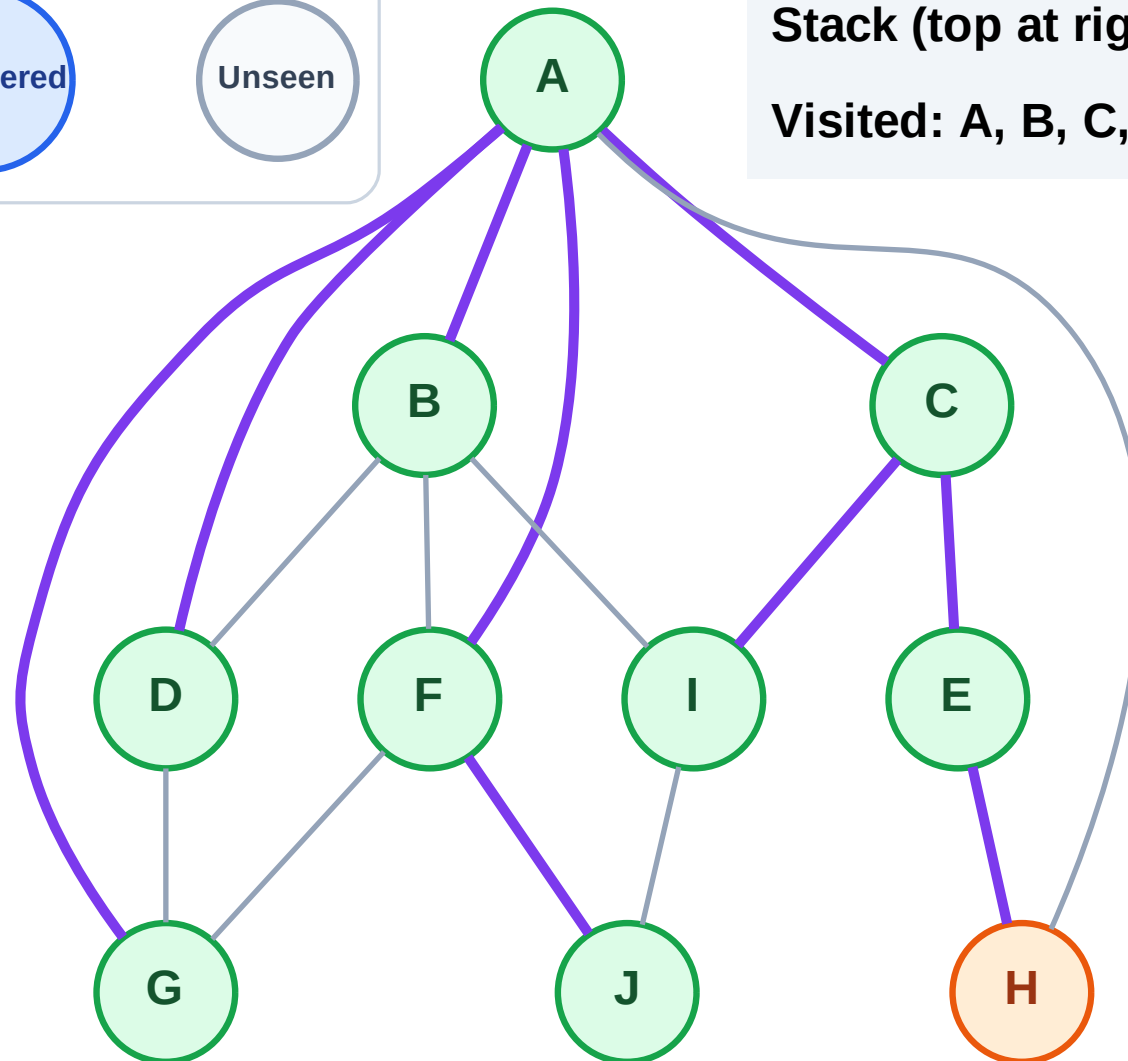
Step 20: Process node H

Step 20: Process node H



Stack (top at right): (empty)

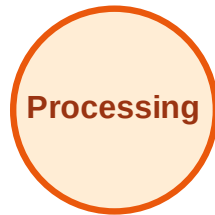
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

